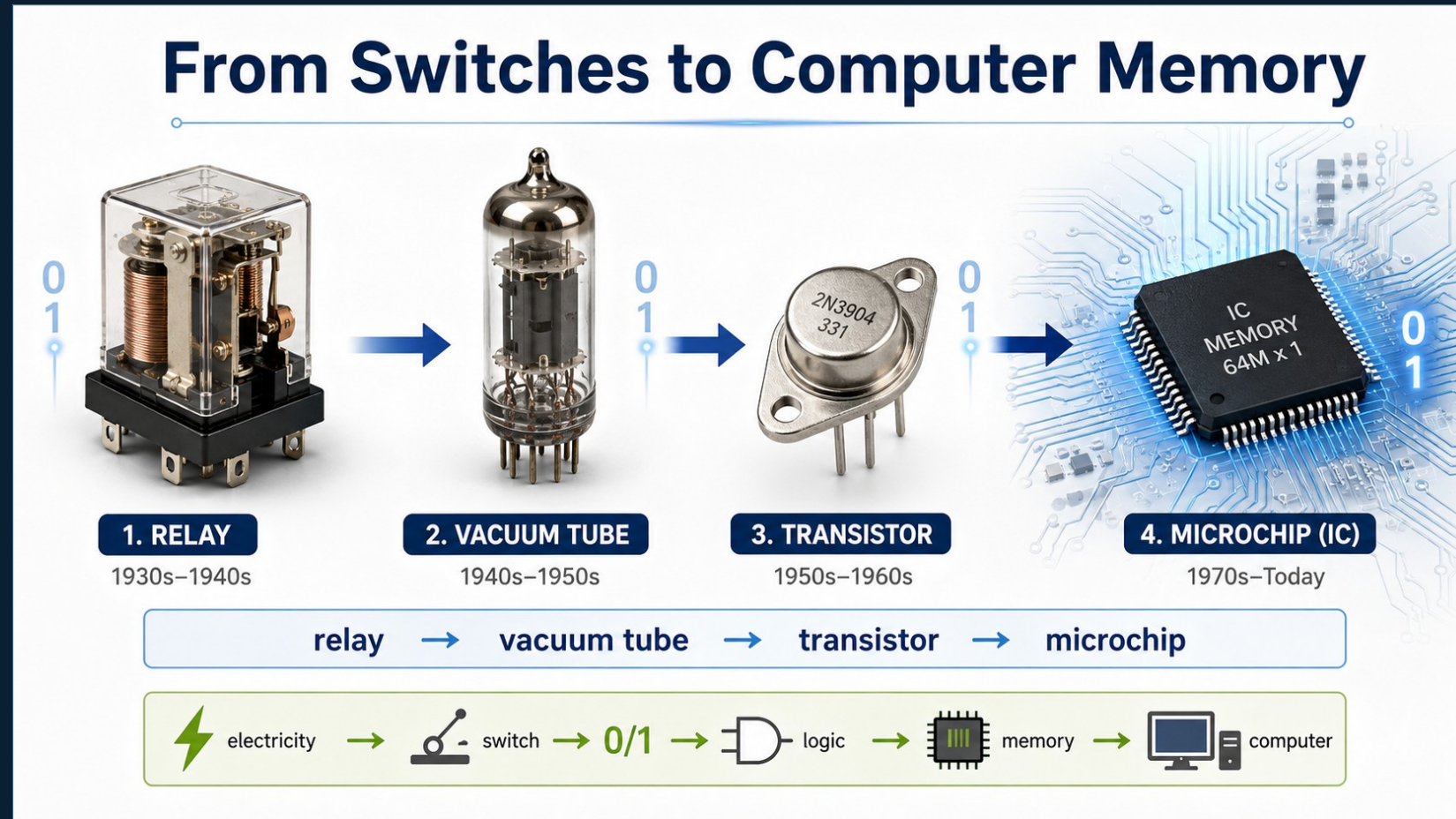


Lecture 9

From Switches to Computer Memory

How electricity became logic, memory, and computers

Main chain: electricity → switch → 0/1 → logic → memory → computer



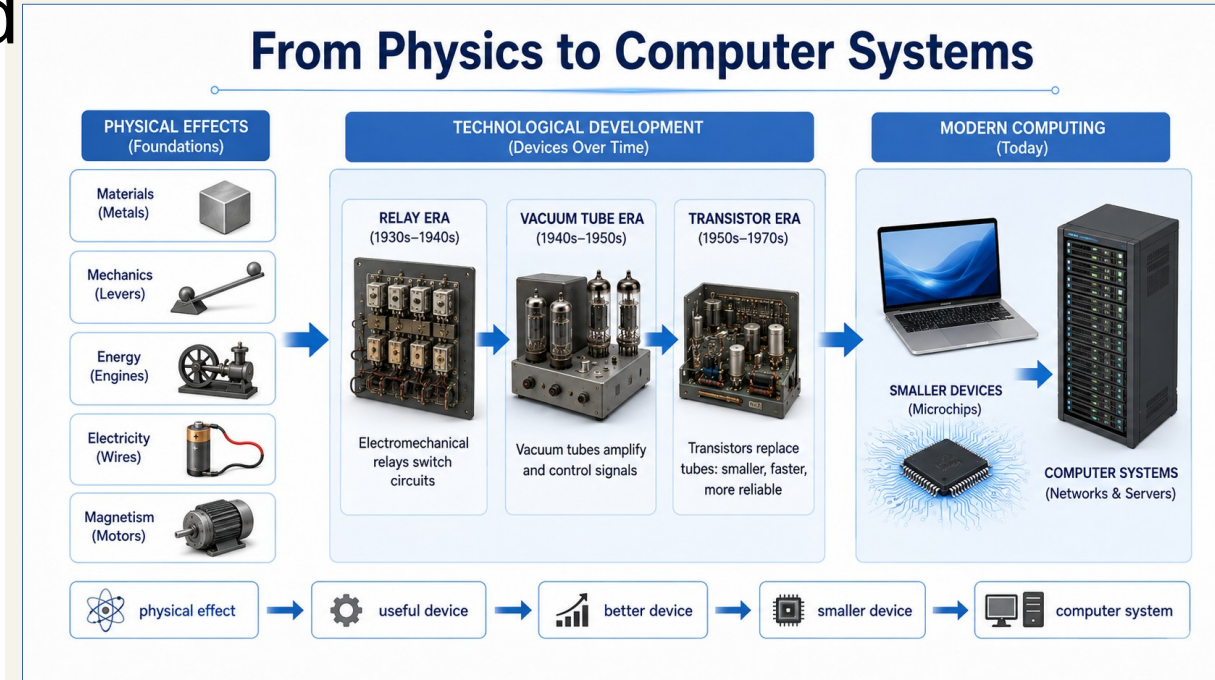
Why this lecture matters

This is the bridge from physics to computer hardware.

In previous lectures we studied materials.

Now we ask a new question:
How did these physical ideas become a computer?

The answer is not magic. It is development step by step:
physical effect → useful device
→ better device → smaller device
→ computer system.



More detail: In previous lectures we studied materials, forces, motion, energy, heat, electricity, magnetism, motors, generators, and big hospital machines.

Use this slide slowly. Students should feel that this lecture is not a new random topic. It is the result of the whole course so far. Physics is not separate from computers. Physics is the foundation that made computers possible.

Course chain so far

Every lecture gave one piece of the computer story.

Materials: What can we build from?

Mechanics: How do parts move and stay stable?

Energy and power: How do machines do work?

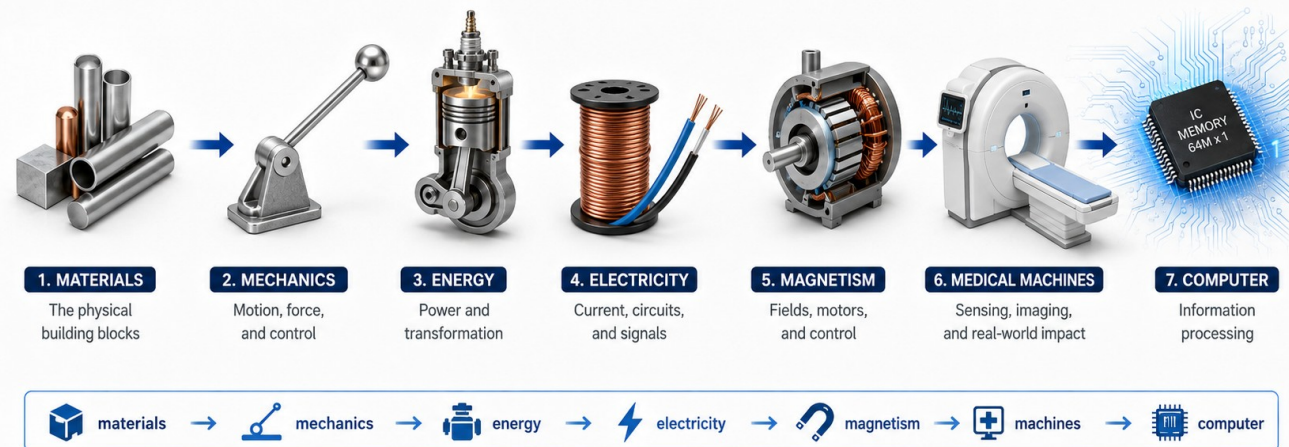
Electricity: How does charge move?

Magnetism: How can current create force?

Hospital machines: How do physical processes become useful output?

Today: How can electricity become logic and memory?

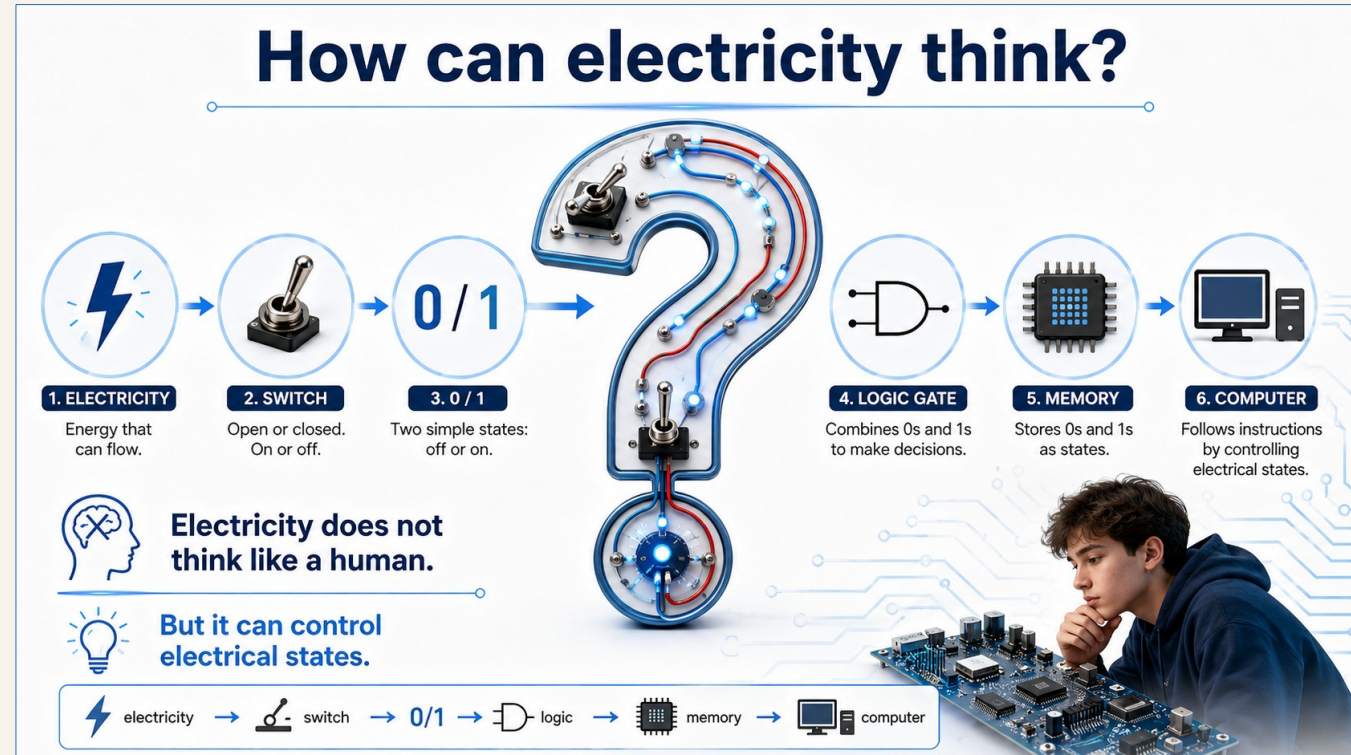
Every lecture gave one piece of the computer story.



Big question

How can electricity "think"?

Careful answer:
Electricity does not think like a human.
But electricity can control switches.
Switches can make 0 and 1.
0 and 1 can make logic.
Logic can calculate.
Memory can store information.

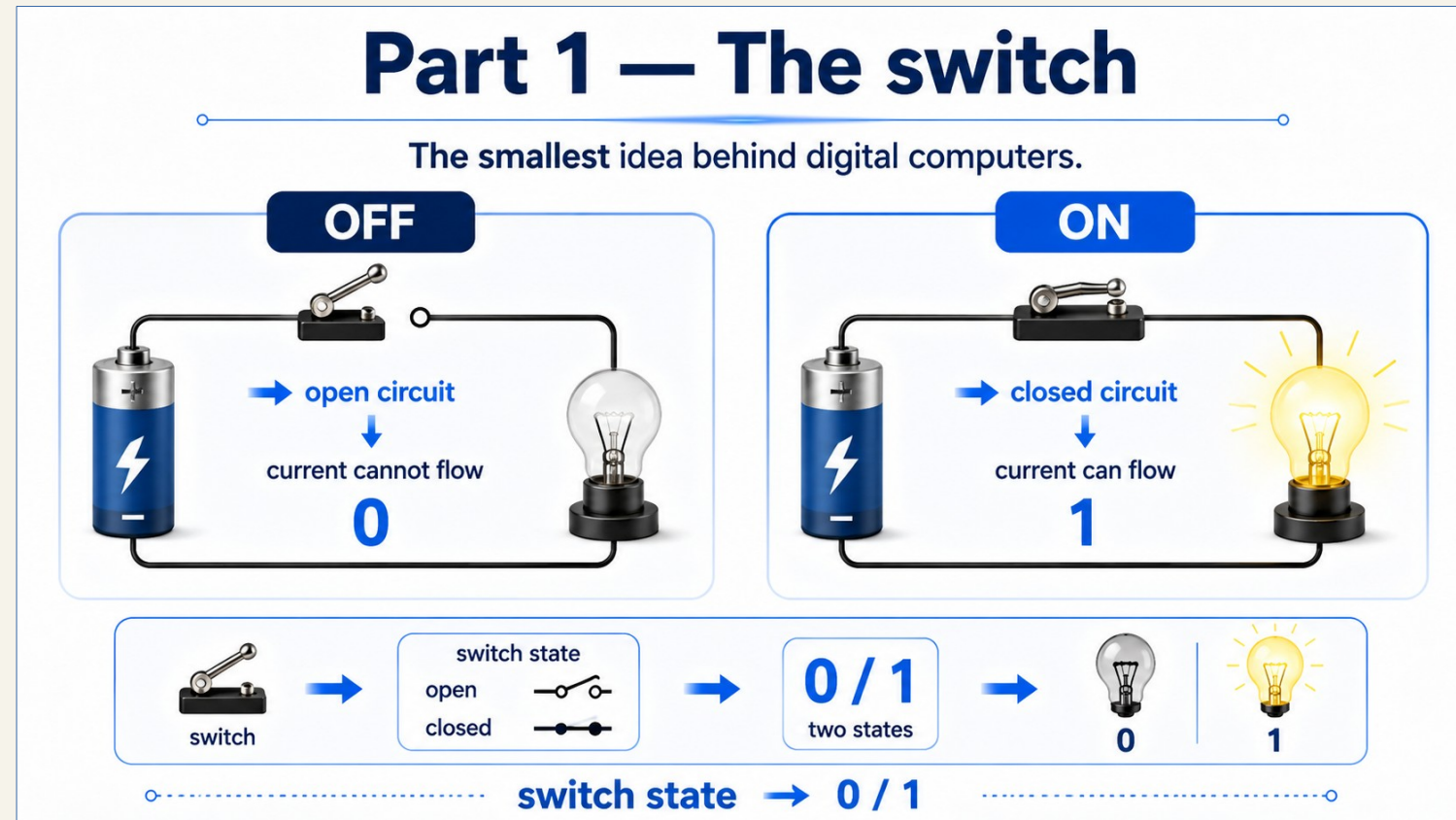


More detail: So the computer is not a magic brain. It is a very large system of controlled electrical states. Do not say the computer understands. Say it follows rules. This is important for future software lessons: software works because hardware can follow exact logical instructions.

Part 1 — The switch

The smallest idea behind digital computers.

A switch is simple.
OFF means the circuit is open.
Current cannot flow.
We can call this state 0.
ON means the circuit is closed.
Current can flow.
We can call this state 1.



More detail: OFF / ON looks too simple. But simple is good for machines.
This is the first real computer idea. Students do not need advanced math yet. They need the feeling that 0 and 1 are physical states, not abstract decoration.

Why computers use two states

Machines need clear differences.

Human language is rich and unclear.

Electric circuits prefer clear states.

Low voltage can mean 0.

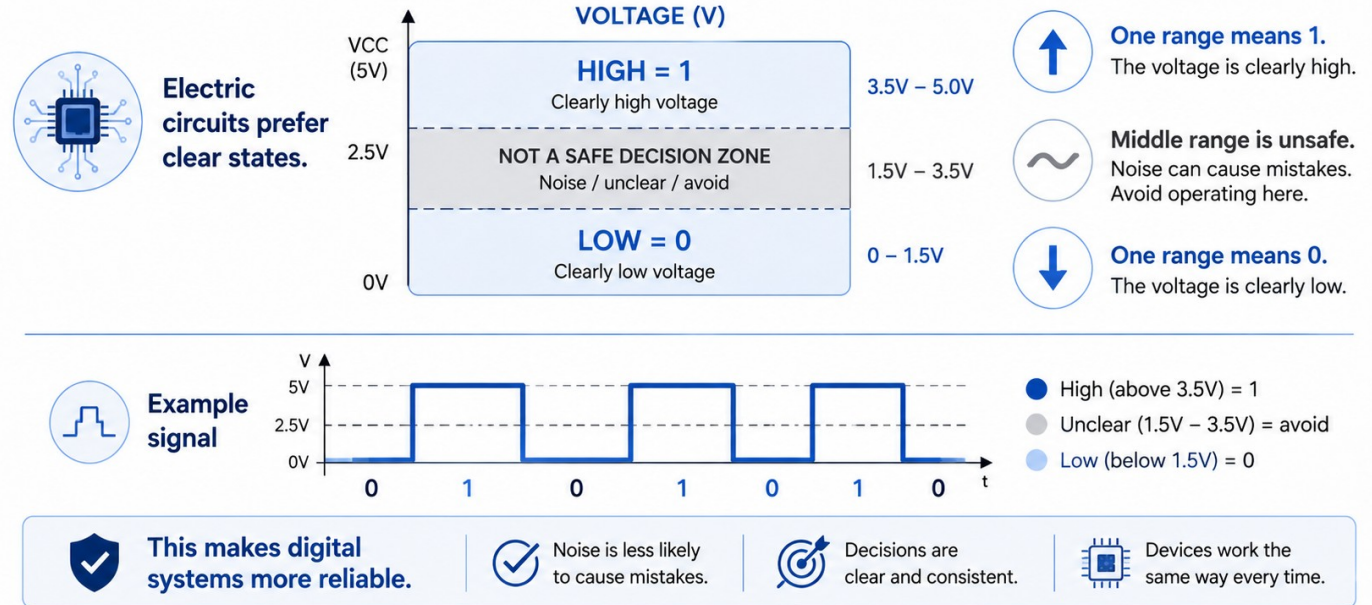
High voltage can mean 1.

The exact voltage can be different in different devices

But the idea is the same:
one range means 0, another range means 1.

Why computers use two states

Machines need clear differences.



More detail: This makes circuits more reliable than trying to use many small levels.

Explain that real electronics do not always use perfect 0 V and perfect 5 V. The important thing is range. This helps students understand why digital systems survive small noise.

Why Computers Use Binary Numbers

A SWITCH HAS TWO STATES

OFF



0

ON



1



- A switch has only two states: OFF and ON.
- A computer uses many tiny switches.
- Because a switch has only two states, a computer uses two symbols: 0 and 1.
- This system is called **binary**.



- A computer must also write numbers using 0 and 1.
- So computer numbers are written in **binary** form.



Big Idea

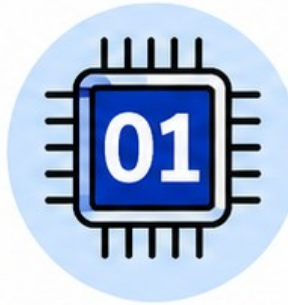
Binary is the natural language of switches.

Decimal and Binary: First Examples

Decimal	Binary
0	0
1	1
2	10
3	11
4	100



Decimal uses
ten symbols:
0 to 9.



Binary uses
only two symbols:
0 and 1.



The number is the same. Only the writing is different.

Binary Place Values: 1, 2, 4, 8

BINARY PLACE VALUES

8

8s place

4

4s place

2

2s place

1

1s place



- Each place can be 0 or 1.
- If the place is 1, we add that value.

WORKED EXAMPLES

0	0	0	1	= 1		0	0	1	0	= 2		0	0	1	1	= 3
8	4	2	1			8	4	2	1			8	4	2	1	



0011 means $0 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1 = 3$.



Binary places grow by powers of two.

Binary Numbers with Switches

Four Switches = 8, 4, 2, 1

8



OFF = 0

4



OFF = 0

2

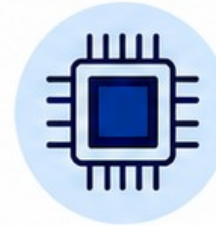


ON = 1

1



ON = 1



Read from left to right.
ON switches add their value.

Worked Example

$$0011 = 0 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1 = 3$$

Example 1:



$$0001 = 1$$

Example 2:



$$0010 = 2$$

Example 3:



$$0011 = 3$$



- Each switch has a value.
- ON (1) adds its value.
- OFF (0) adds nothing.
- Add the values of all ON switches.



Computer numbers can be shown by many ON/OFF switches.

Simple Binary Addition with Switches

Example 1

$$\begin{array}{r} 01 \\ + 01 \\ \hline 10 \end{array}$$

2s place



1s place



1 + 1 does not fit in one place.
We write 0 and carry 1 to the next place.

Example 2

$$\begin{array}{r} 010 \\ + 010 \\ \hline 100 \end{array}$$

4s place



2s place



1s place



2 + 2 = 4.

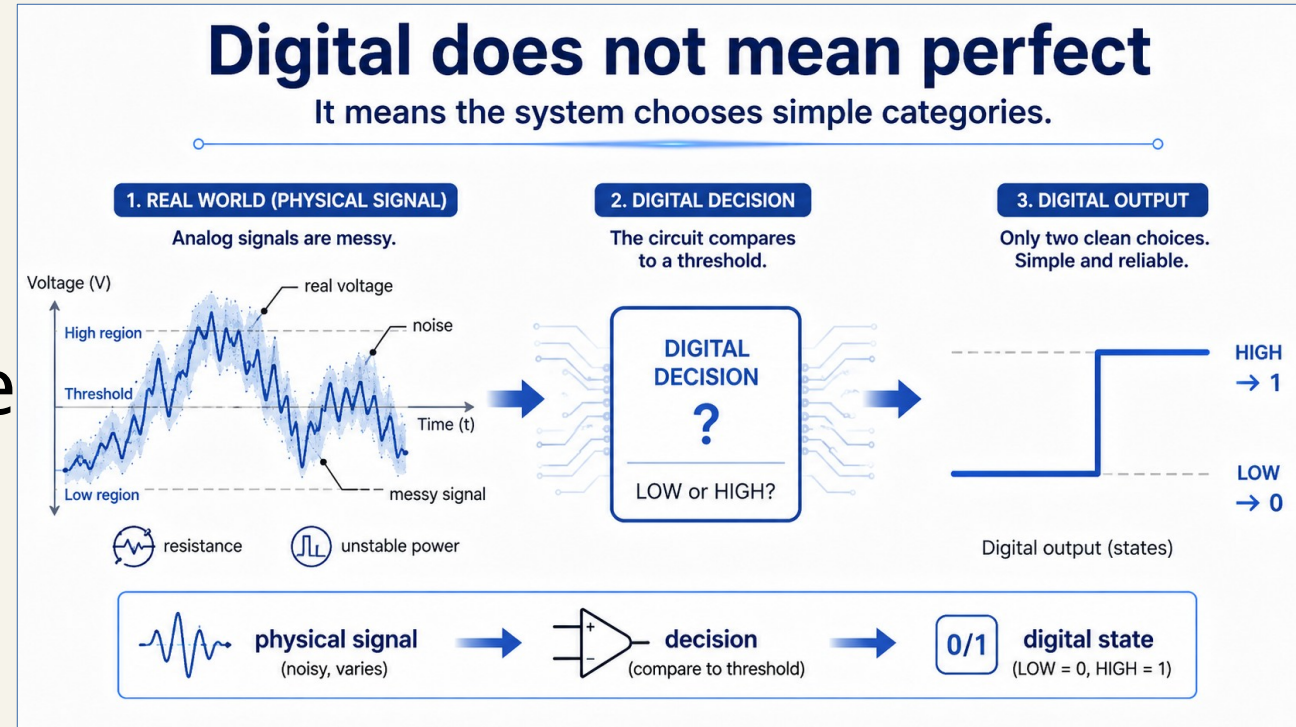


Computers add using 0 and 1.

Digital does not mean “perfect”

It means the system chooses simple categories.

Real voltage can be a little messy.
Wires have resistance.
Signals can have noise.
Power can be unstable.
A digital circuit asks a simple question:
Is this signal LOW or HIGH?
Then it decides:



More detail: LOW → 0 HIGH → 1

This is a useful future idea. When computers fail, the problem is often physical: bad power, damaged cable, heat, corrosion, or weak signal. Digital systems still live in the physical world.

From one switch to a rule

One switch gives one bit. Two switches can make logic.

One switch can show one state: 0 or 1.

Two switches can make a rule.

Example:

A lamp turns ON only when switch A is ON and switch B is ON.

This is already computer logic.

The computer follows a rule exactly.

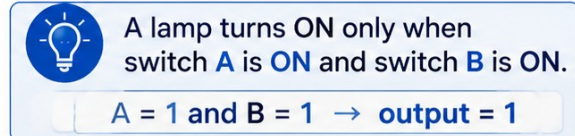
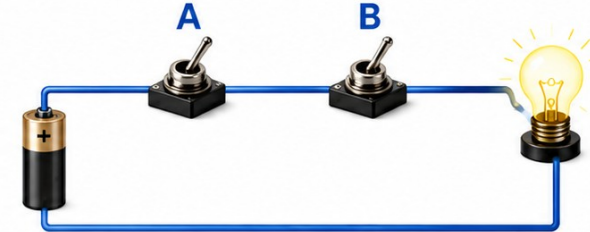
From one switch to a rule

One switch gives one bit. Two switches can make logic.

ONE SWITCH = ONE BIT



TWO SWITCHES IN SERIES (AND RULE)



TRUTH TABLE (AND)		
A	B	OUTPUT (Lamp)
0	0	0
0	1	0
1	0	0
1	1	1



Two switches in series follow the AND rule: the output is 1 only when both inputs are 1.

Use a very physical example. Two people must both agree; two switches must both be ON. This becomes the AND rule.

AND logic

A strict rule: both must be true.

AND means:
A and B must both be 1.

$0 \text{ AND } 0 = 0$

$0 \text{ AND } 1 = 0$

$1 \text{ AND } 0 = 0$

$1 \text{ AND } 1 = 1$

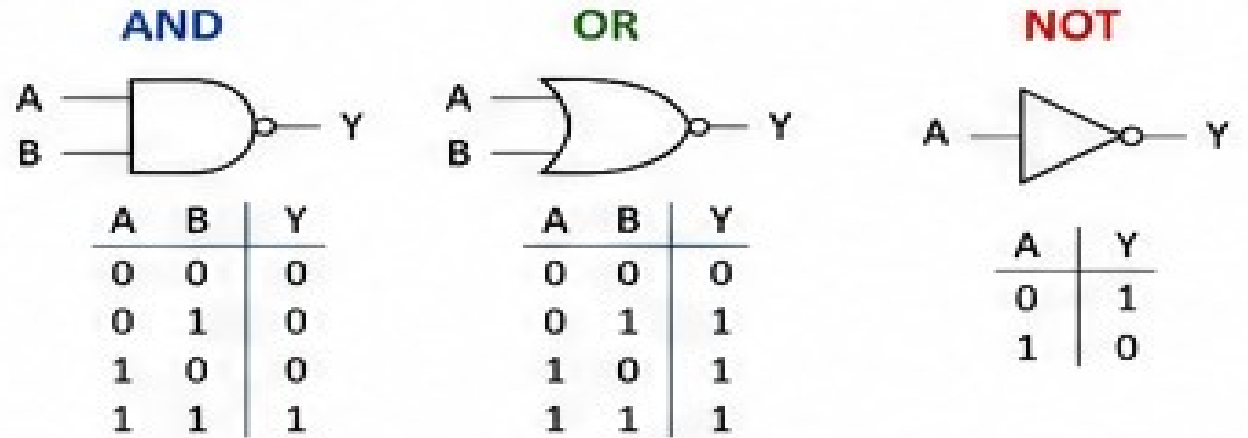
Human example:

33. Logic Gates – The Building Blocks

Computers use simple logic gates.

These gates work with 0 and 1.

They are made from transistors.



More detail: The door opens only if the card is valid AND the door is not locked.

Say the truth table slowly. This is hard for weak students, but it is the birth of programming logic. They will meet AND again in software, databases, and hospital information systems.

OR logic

A more flexible rule: one is enough.

OR means:

If A is 1 or B is 1,
the output is 1.

$0 \text{ OR } 0 = 0$

$0 \text{ OR } 1 = 1$

$1 \text{ OR } 0 = 1$

$1 \text{ OR } 1 = 1$

Hospital example:

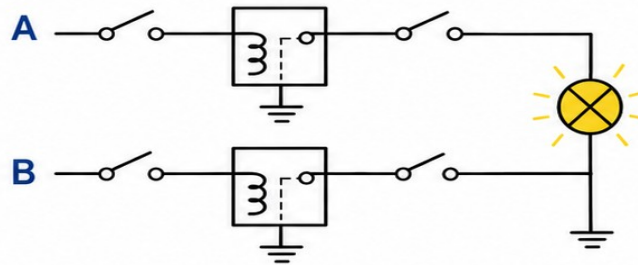
Relays Can Do Logic

4

Relays can be combined to perform logic operations.

AND gate with relays

Both inputs A and B must be ON
for the output lamp to turn ON.



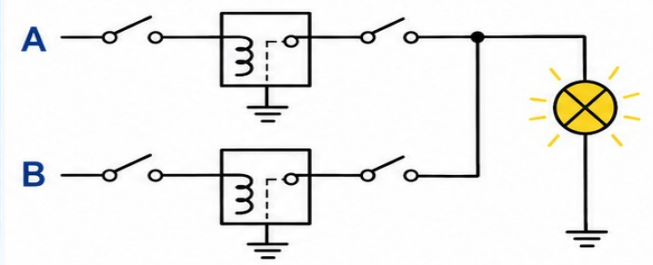
A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1



Both A and B must be 1.

OR gate with relays

Either input A or B (or both) can
turn the output lamp ON.



A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1



Either A or B (or both) can be 1.

More detail: If the patient has strong pain OR high fever, staff must check the patient.
This slide connects computer logic to hospital thinking. OR is not just math. It is a rule for action.

NOT logic

A reverse rule.

NOT changes the signal.

NOT 0 = 1

NOT 1 = 0

Human example:

If the test is NOT
complete, do not archive
the file.

Computer systems use
NOT all the time:
not ready, not connected,
not allowed, not found.

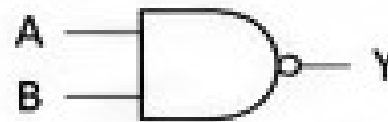
33. Logic Gates – The Building Blocks

Computers use simple logic gates.

These gates work with 0 and 1.

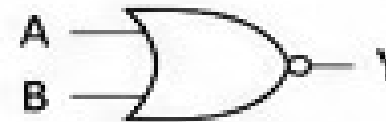
They are made from transistors.

AND



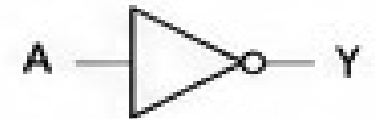
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT



A	Y
0	1
1	0

This prepares students for later conditions in software: if not connected, if not paid, if not found.

Logic gates are physical devices

Logic is not only an idea in a textbook.

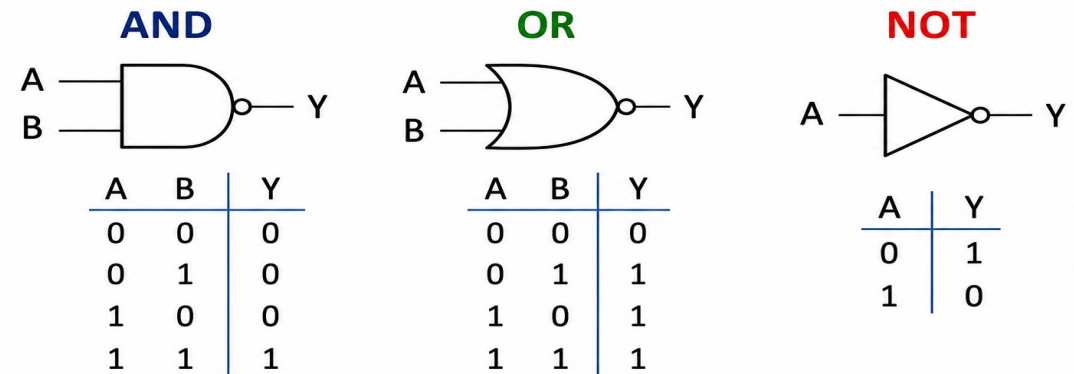
A logic gate is a physical circuit.
It receives electrical inputs.
It follows a simple rule.
It gives an electrical output.
Input A and input B are not words.
They are real voltages in wires.
This is where physics becomes computer logic.

33. Logic Gates – The Building Blocks

Computers use simple logic gates.

These gates work with 0 and 1.

They are made from transistors.



Keep repeating: logic is physical. This is the important bridge. Later, software looks abstract, but at the bottom there are physical signals.

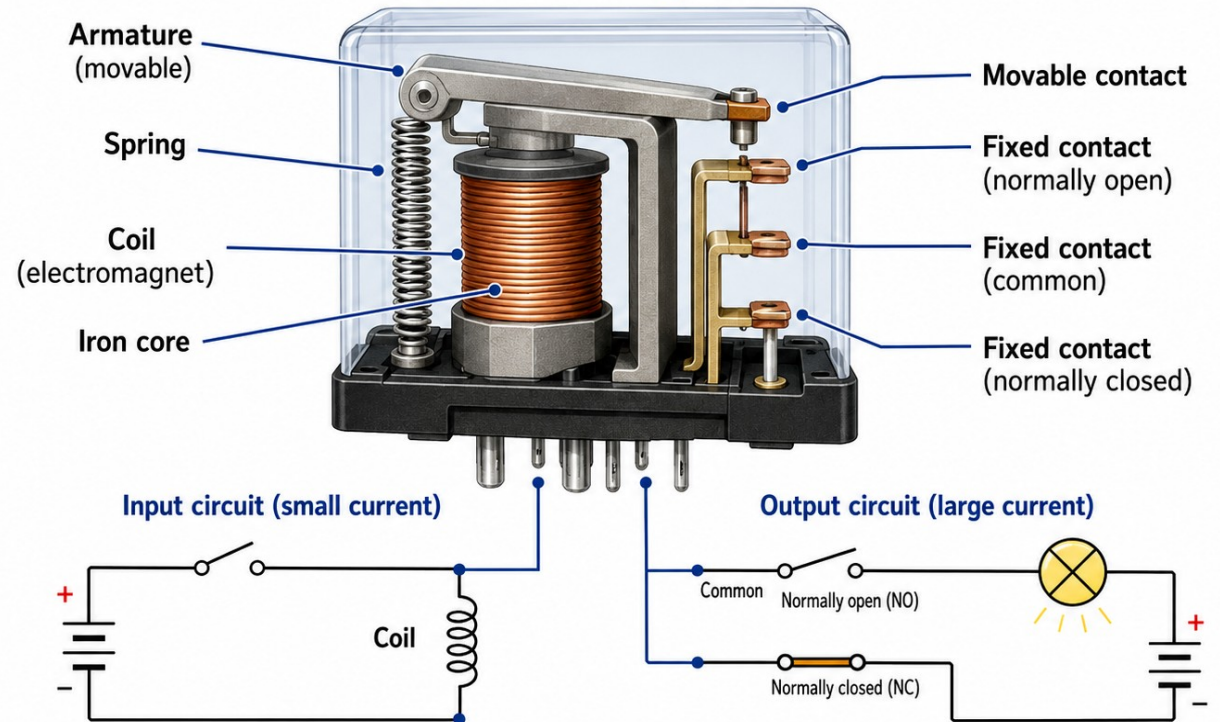
Part 2 — Relay

The first big bridge: electricity controls electricity.

A human can press a switch with a finger.
But can electricity press a switch?
Yes.
A relay does this.
A relay uses electromagnetism.
Small current in one circuit can control another circuit.

Relay — Basic Parts

A relay is an electrically controlled switch.



A small current in the coil controls a larger current in the output circuit.

This is where Lecture 7 becomes useful. Current can make a magnetic field. A magnetic field can move metal. That motion can close another circuit.

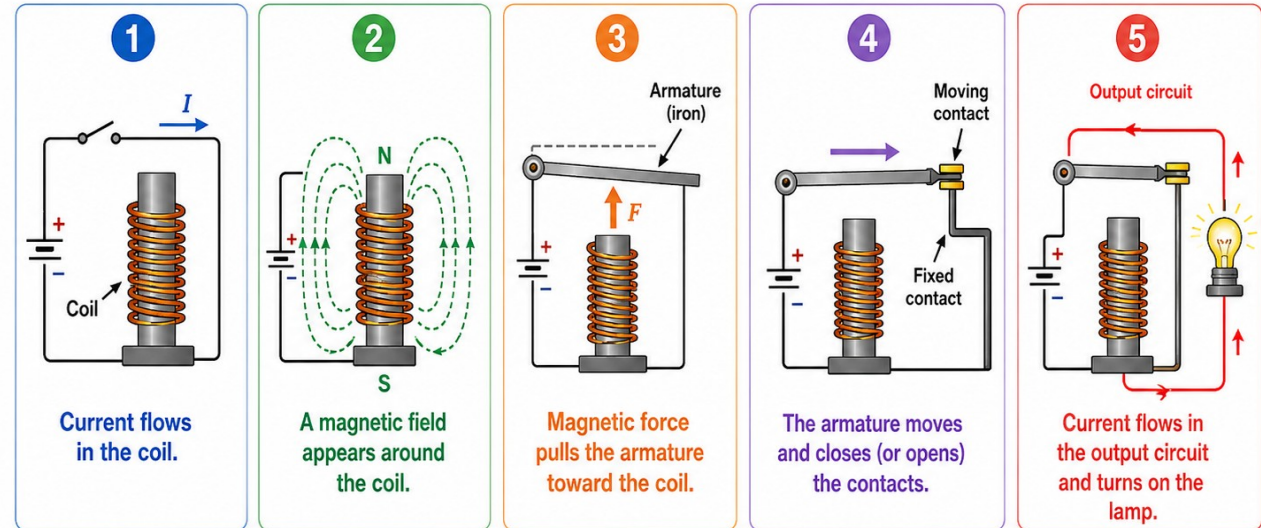
Relay physics

From current to magnetic force.

A relay has a coil of wire.
When current flows in the coil,
the coil becomes an
electromagnet.
The electromagnet pulls a
metal armature.
The armature moves a contact.
The contact opens or closes
another circuit.
Chain:
current → magnetic field →
force → moving contact →
output current

Relay — How It Works (Physics Chain)

1. Current → 2. Magnetic field → 3. Force → 4. Moving contact → 5. Output current



Cause → Effect: Current → Magnetic field → Force → Moving contact → Output current

This slide should be visually strong later. It is a perfect example of physics becoming technology.

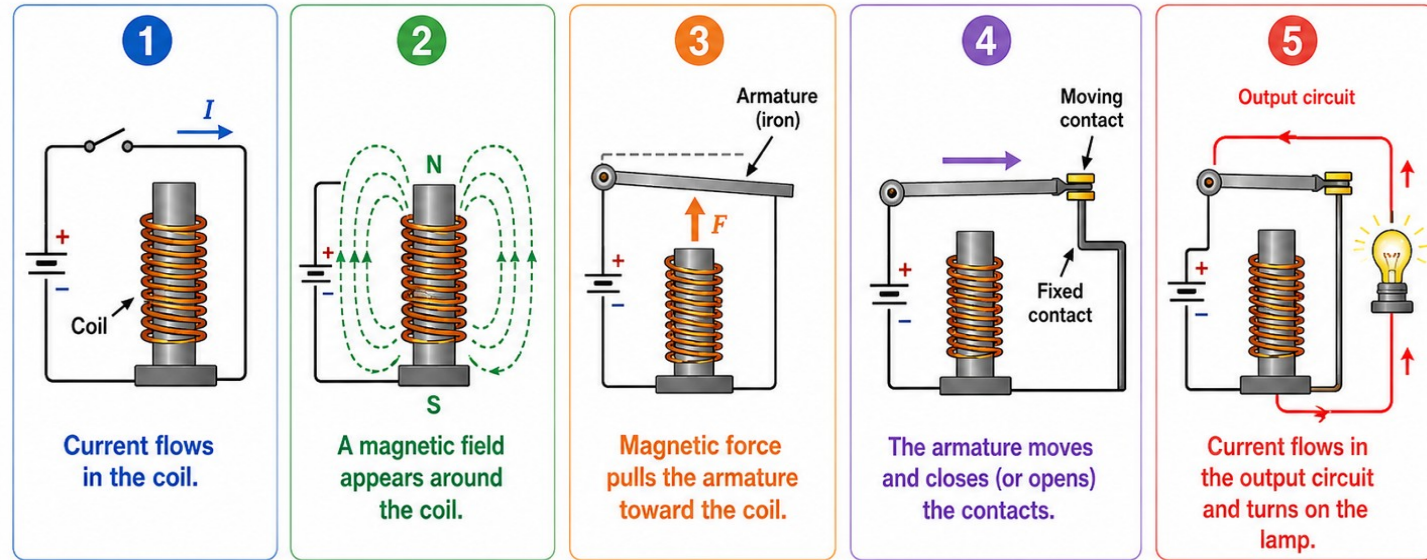
Relay step by step

A machine can switch without a human finger.

1. Input current flows in the coil.
 2. Coil creates a magnetic field.
 3. Magnetic field pulls metal.
 4. Metal contact moves.
 5. Output circuit closes.
 6. Lamp, motor, or another circuit turns ON.
- This is control.

Relay – How It Works (Physics Chain)

1. Current → 2. Magnetic field → 3. Force → 4. Moving contact → 5. Output current



Cause → Effect: Current → Magnetic field → Force → Moving contact → Output current

Why relay was revolutionary

Electricity could control a decision.

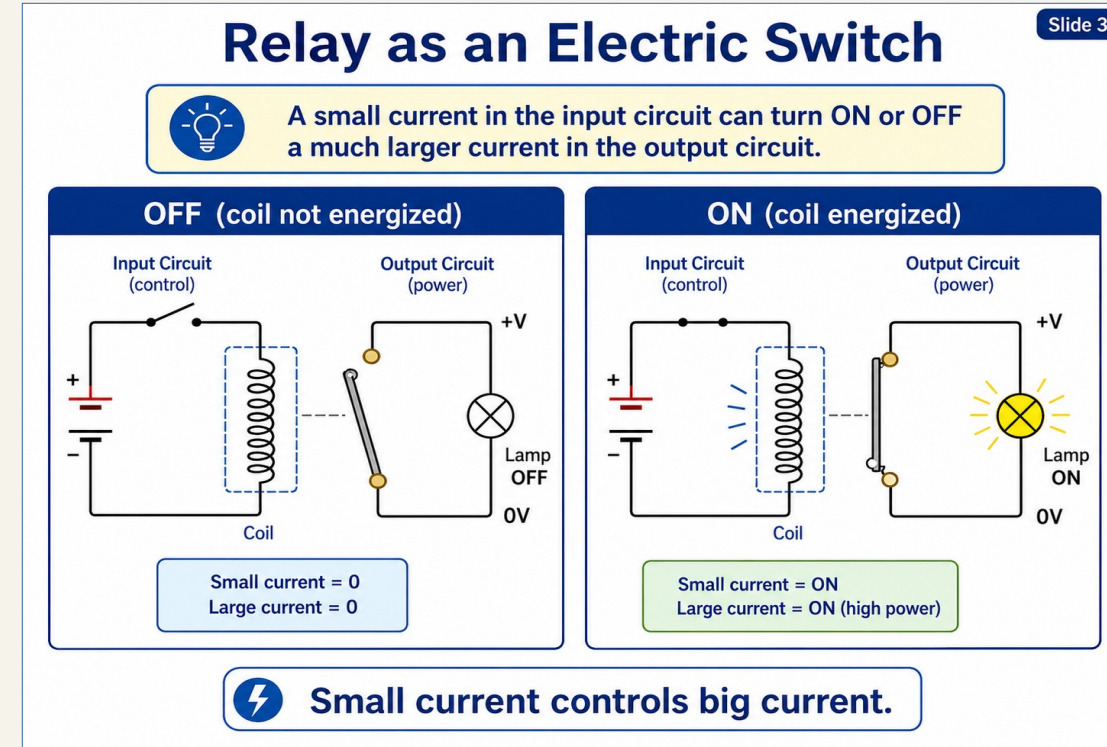
Before relay logic, many machines were mechanical.

Gears, levers, wheels, shafts, and human operation were common.

A relay made a new idea practical: one electrical signal can control another electrical signal.

That means circuits can make decisions automatically.

This was one road toward automatic computing.



Say "one road," not "the only road." Computer history is complex. But for this course, relay is a clean bridge from physics to logic.

Relay AND logic

Logic can be built with physical contacts.

Imagine two relay contacts in series. If contact A is open, current cannot pass. If contact B is open, current cannot pass. Only if A and B are closed, current passes. This is AND logic made from metal contacts. Logic rule = physical

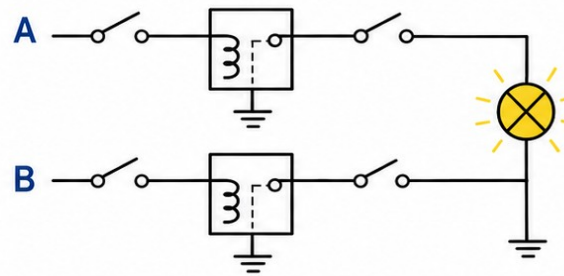
Relays Can Do Logic

4

Relays can be combined to perform logic operations.

AND gate with relays

Both inputs A and B must be ON for the output lamp to turn ON.



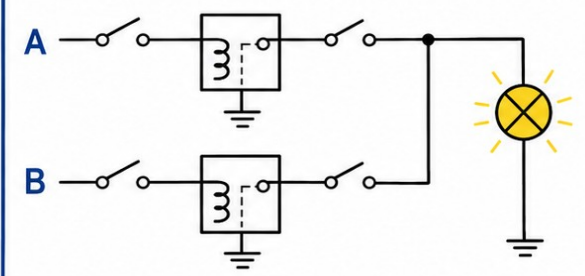
A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1



Both A and B must be 1.

OR gate with relays

Either input A or B (or both) can turn the output lamp ON.



A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1



Either A or B (or both) can be 1.

This is a key slide. It shows that truth tables are not only classroom symbols. They can be built from physical devices.

Relay OR logic

Parallel paths make OR.

Now imagine two contacts in parallel.
Current can pass through path A.
Or current can pass through path B.
If either path is closed, the output turns ON.
This is OR logic made from metal contacts.
Different wiring gives different logic.

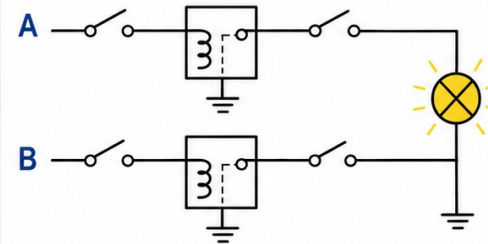
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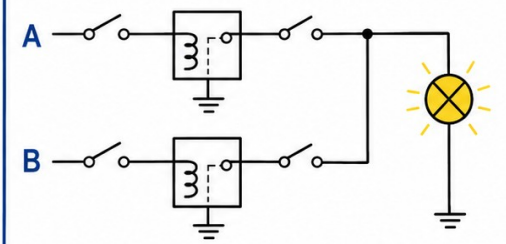
A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1



Both A and B must be 1.

OR gate with relays

Either input A or B (or both) can turn the output lamp ON.



A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1



Either A or B (or both) can be 1.

After this slide, students should understand that wiring is a kind of programming. In old machines, changing wiring could change what the machine did.

Historical bridge: relay computers

Relays were slow, but they were real computers.

Important correction:
Relay computers were not a joke and not simply failed machines.
They were real steps in computer history.
They used many relays to represent numbers and operations.
But they were limited: mechanical movement, noise, wear, slow speed, large size.

41. Evolution of Computers (Summary)

Mechanical Calculators	Relay Computers	Vacuum Tube Computers	Transistor Computers	Integrated Circuits	Modern Computers
					
1600s–1800s	1930s–1940s	1940s–1950s	1950s–1960s	1970s	Today
manual mechanical devices	relays, slow, noisy	faster, but big, hot, fragile	smaller, faster, more reliable	many components on one chip	very small, very fast, powerful

Be honest: relay machines worked. The problem was not “they did not work.” The problem was that mechanical switching limited speed, size, and reliability.

Example: Zuse Z3

A working electromechanical computer from 1941.

Konrad Zuse built the Z3 in Germany in 1941.
It used thousands of relays.
It worked with binary arithmetic.
It was programmable using punched film.
For our lecture, the important idea is simple:
relay + binary states + program control = a real step toward modern computers.

42. Real Example: Old Relay Computer (Zuse Z3)



Photo: reconstructed Zuse Z3 in the Deutsches Museum.

- Built in 1941.
- Used about 2,600 relays.
- Could do calculations with binary numbers.
- Very slow, about 1 operation every few seconds.
- But it was a real, working computer.

Computer History Museum describes the Z3 as an early computer using 2,300 relays and floating-point binary arithmetic. Use this as a careful historical example, not as a long museum discussion.

Example: Harvard Mark I

Large electromechanical automation.

The Harvard Mark I / IBM ASCC was another famous electromechanical machine. It combined switches, relays, counters, rotating parts, and long wiring. It could run long computations automatically. This shows a common pattern in technology: use the best available physical devices, then improve the weak parts.

41. Evolution of Computers (Summary)

Mechanical Calculators	Relay Computers	Vacuum Tube Computers	Transistor Computers	Integrated Circuits	Modern Computers
					
1600s–1800s	1930s–1940s	1940s–1950s	1950s–1960s	1970s	Today
manual mechanical devices	relays, slow, noisy	faster, but big, hot, fragile	smaller, faster, more reliable	many components on one chip	very small, very fast, powerful

This is not for memorizing dates. The goal is to see scale: early computers were room-sized physical systems, not invisible apps.

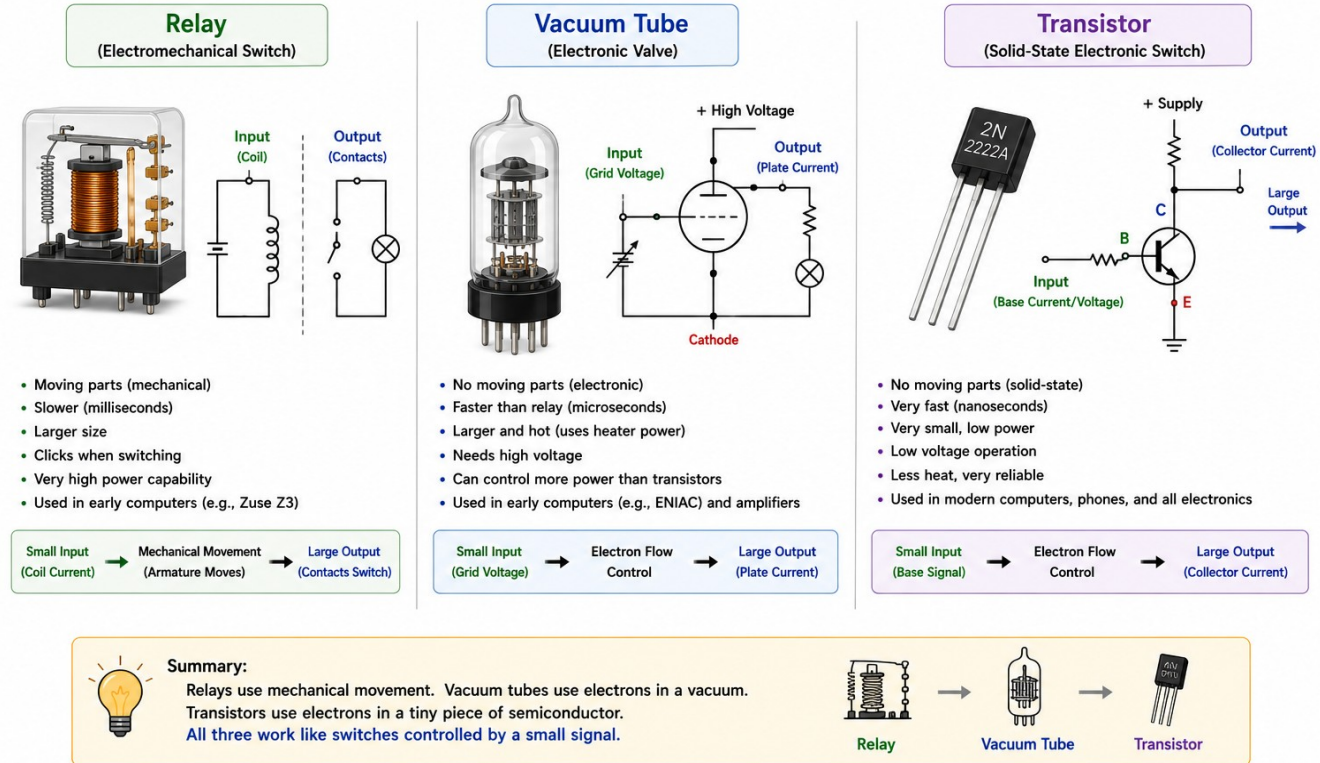
The relay problem

Mechanical movement is the bottleneck.

A relay must move metal.
Moving metal takes time.
Moving parts can wear out.
Contacts can become dirty or damaged.
The machine makes sound and heat.
So engineers asked:
Can we control current without moving parts?

7. Relay vs Vacuum Tube vs Transistor

The same idea: a small input controls a larger output. *Technology gets smaller, faster, cooler, and more reliable.*



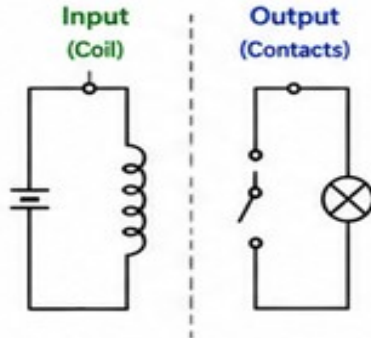
This is the transition to vacuum tubes. The question comes naturally from physics: if movement is slow, remove movement.

7. Relay vs Vacuum Tube vs Transistor

The same idea: a small input controls a larger output. Technology gets smaller, faster, cooler, and more reliable.

Relay

(Electromechanical Switch)

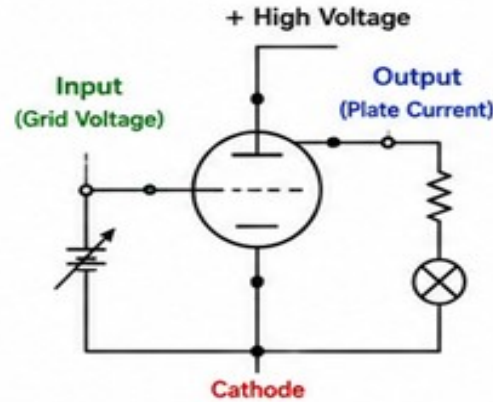


- Moving parts (mechanical)
- Slower (milliseconds)
- Larger size
- Clicks when switching
- Very high power capability
- Used in early computers (e.g., Zuse Z3)

Small Input (Coil Current) → Mechanical Movement (Armature Moves) → Large Output (Contacts Switch)

Vacuum Tube

(Electronic Valve)

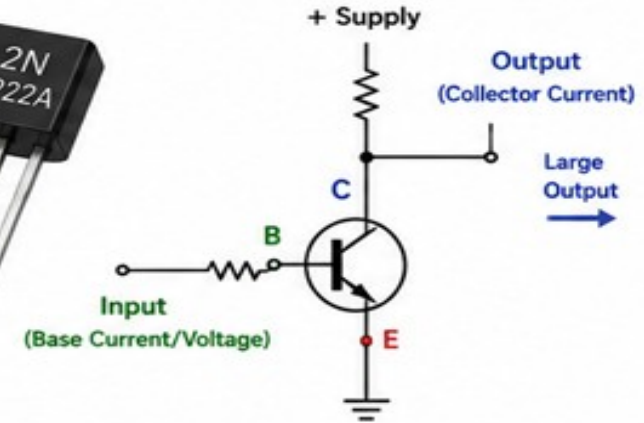


- No moving parts (electronic)
- Faster than relay (microseconds)
- Larger and hot (uses heater power)
- Needs high voltage
- Can control more power than transistors
- Used in early computers (e.g., ENIAC) and amplifiers

Small Input (Grid Voltage) → Electron Flow Control → Large Output (Plate Current)

Transistor

(Solid-State Electronic Switch)



- No moving parts (solid-state)
- Very fast (nanoseconds)
- Very small, low power
- Low voltage operation
- Less heat, very reliable
- Used in modern computers, phones, and all electronics

Small Input (Base Signal) → Electron Flow Control → Large Output (Collector Current)



Summary:

Relays use mechanical movement. Vacuum tubes use electrons in a vacuum. Transistors use electrons in a tiny piece of semiconductor. All three work like switches controlled by a small signal.



Relay



Vacuum Tube



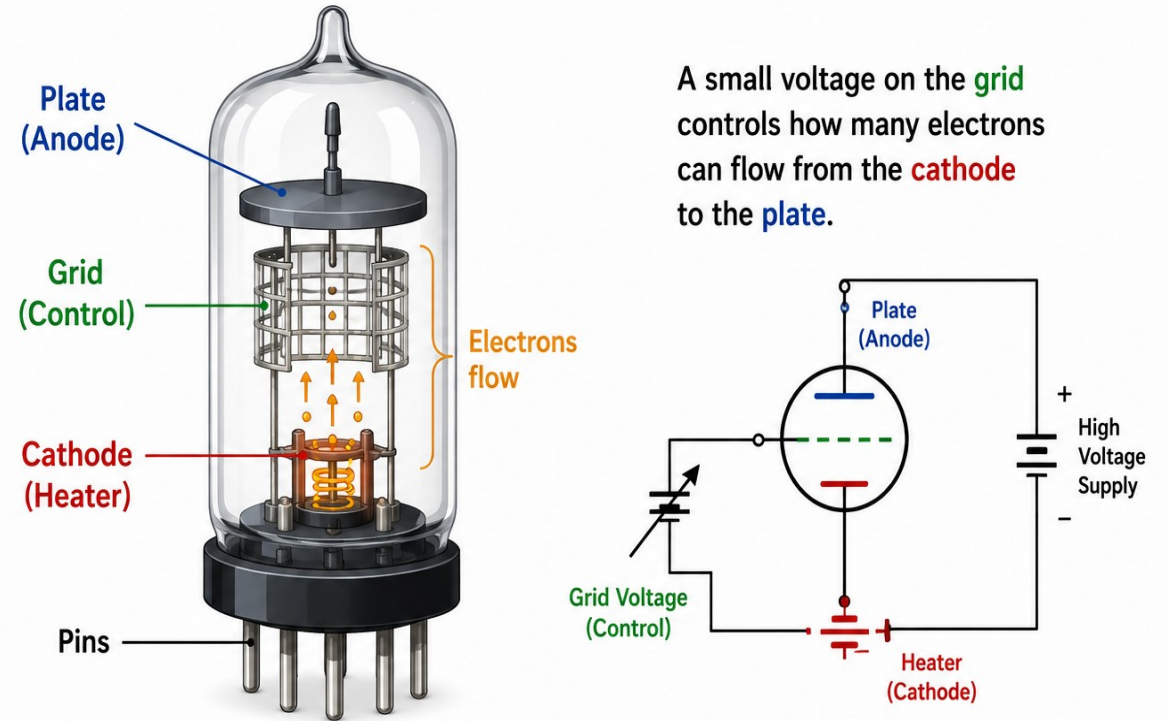
Transistor

Part 3 — Vacuum tubes

Control electron flow without moving metal contacts.

A vacuum tube controls electrons inside a sealed glass tube.
It can act like a switch or amplifier.
It has no moving relay contact.
So it can switch faster than a relay.
But it has problems:
large size, heat, fragility,
high power use.

5. Vacuum Tube Triode – Principle



A small voltage on the **grid** controls how many electrons can flow from the **cathode** to the **plate**.

The **grid** acts like a gate. A small change in **grid voltage** can strongly change the **plate current**.

Do not explain every electrode. Keep it conceptual: electrons move in vacuum; an electric field can control their flow. That is enough for this level.

Vacuum tube physics

Electric field controls moving electrons.

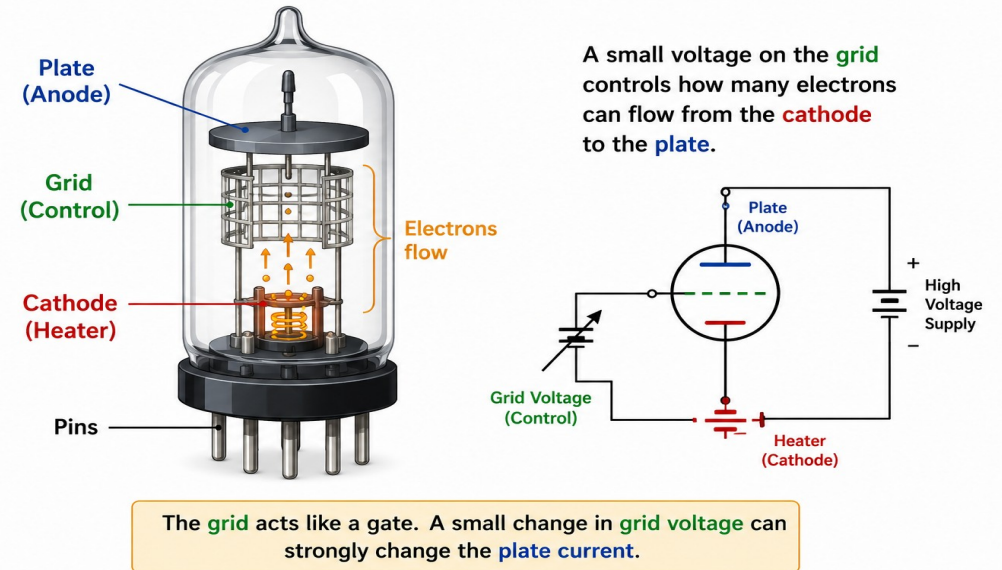
Inside a vacuum tube, electrons can travel through empty space. A hot cathode can release electrons.

An anode can attract electrons. A control grid can change the flow.

Small change in control can make a large change in current. This is another form of control.

This slide preserves the physics. We are not just saying “old part.” We are explaining why it was technologically useful: controlling electron flow.

5. Vacuum Tube Triode – Principle



Electronic computers

Faster switching changed what computers could do.

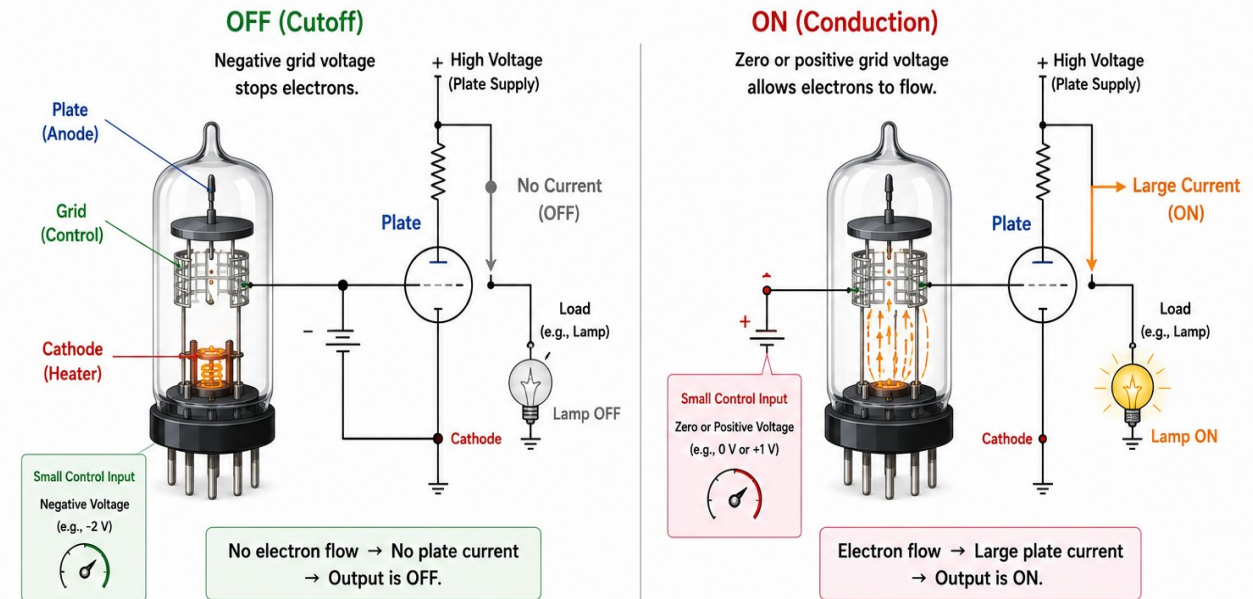
When switches became electronic, computers became much faster. Electronic switching removed the slow mechanical contact. This allowed more operations per second. But early electronic computers were still huge. They used many vacuum tubes and a lot of power. They needed maintenance and cooling.

This connects directly to hospital IT: large systems need power, cooling, maintenance, and safe operation. Computers are still machines.

6. Vacuum Tube as an Electronic Switch

A vacuum tube can act like a switch.

A **small** control signal on the grid controls a **much larger** current between the plate and the cathode.



Key Idea: A **tiny control voltage** on the grid controls a **large current** in the plate circuit. Vacuum tubes were used in early computers and amplifiers for this reason.

Example: ENIAC

A famous early electronic computer.

ENIAC was built in the 1940s in the United States. It used many vacuum tubes. It was much faster than electromechanical machines. It was also huge, hot, and difficult to maintain. Lesson: faster physics gave better computing, but created new engineering problems.

Computer History Museum notes ENIAC used almost 18,000 tubes and that tube failures were a maintenance challenge. This is a perfect example of progress plus new problems.

43. Real Example: Vacuum Tube Computer (ENIAC)

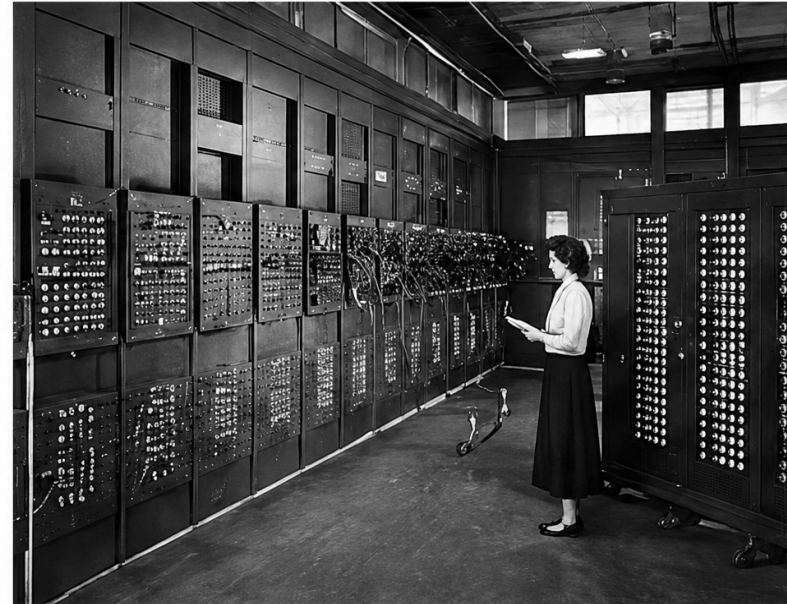


Photo: ENIAC at the University of Pennsylvania.

- Built in 1945.
- Used about 18,000 vacuum tubes.
- Much faster than relay computers.
- Very large room, very hot.
- Used for science and military calculations.

Technology development pattern

Every solution creates the next problem.

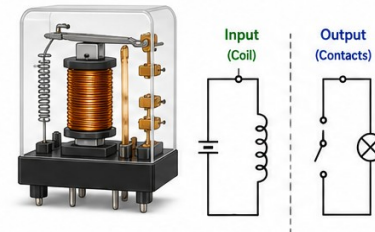
Relay solved automatic electrical control.
But relay was slow and mechanical.
Vacuum tube solved fast electronic switching.
But vacuum tube was big, hot, and fragile.
So the next question became:
Can we make an electronic switch that is small, cool, and reliable?

7. Relay vs Vacuum Tube vs Transistor

The same idea: a small input controls a larger output. Technology gets smaller, faster, cooler, and more reliable.

Relay

(Electromechanical Switch)

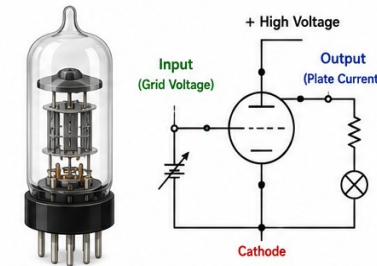


- Moving parts (mechanical)
- Slower (milliseconds)
- Larger size
- Clicks when switching
- Very high power capability
- Used in early computers (e.g., Zuse Z3)

Small Input (Coil Current) → Mechanical Movement (Armature Moves) → Large Output (Contacts Switch)

Vacuum Tube

(Electronic Valve)

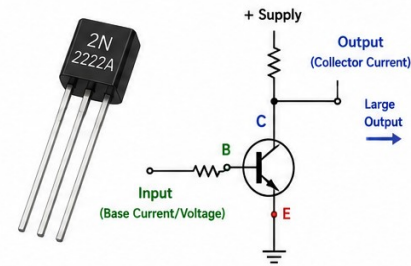


- No moving parts (electronic)
- Faster than relay (microseconds)
- Larger and hot (uses heater power)
- Needs high voltage
- Can control more power than transistors
- Used in early computers (e.g., ENIAC) and amplifiers

Small Input (Grid Voltage) → Electron Flow Control → Large Output (Plate Current)

Transistor

(Solid-State Electronic Switch)



- No moving parts (solid-state)
- Very fast (nanoseconds)
- Very small, low power
- Low voltage operation
- Less heat, very reliable
- Used in modern computers, phones, and all electronics

Small Input (Base Signal) → Electron Flow Control → Large Output (Collector Current)



Summary:

Relays use mechanical movement. Vacuum tubes use electrons in a vacuum. Transistors use electrons in a tiny piece of semiconductor. All three work like switches controlled by a small signal.



Relay



Vacuum Tube



Transistor

This is the historical story the user wanted: physics and history together. Technology is not random; each generation solves the weakness of the previous one.

Part 4 — Transistor

The tiny electronic switch.

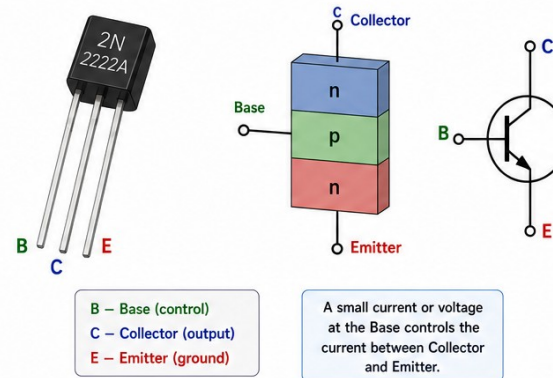
A transistor can control current in a semiconductor. It can work as a switch:
OFF = current blocked = 0
ON = current allowed = 1
It has no moving parts.
It is small, fast, and reliable compared with relays and tubes.
This changed computers completely.

8. Transistor as a Tiny Electronic Switch

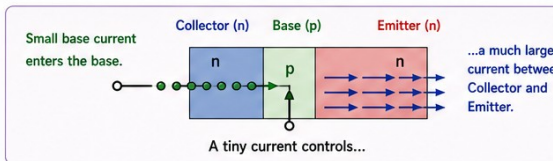


A transistor uses a very **small input** signal to control a **larger current**. It is the basic building block of modern electronics and computers.

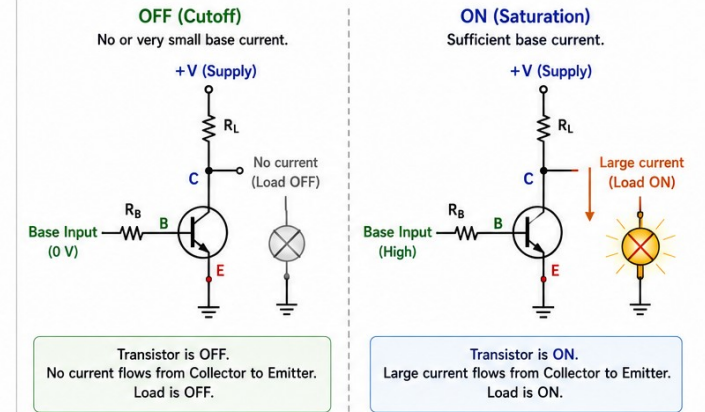
NPN Transistor – Structure and Symbol



How It Works (Idea)



Operation as a Switch (NPN Transistor)



Key Points

- Very **small** input controls **large** output.
- Very fast (nanoseconds).
- Very small and low power.
- No moving parts (solid-state).
- Billions are used inside modern chips.



Used In

- Computers (CPUs, memory, GPUs)
- Phones, tablets, IoT devices
- Power supplies and chargers
- Cars, appliances, almost everything



Keep the transistor conceptual. The deep semiconductor physics can wait. For now, transistor equals tiny controlled switch.

Transistor physics, simplified

Semiconductors can be controlled.

A semiconductor is not a perfect conductor and not a perfect insulator.

Its ability to conduct can be controlled.

A small signal can allow or block a larger current path.

That is why a transistor can work as a switch.

Same big idea again:

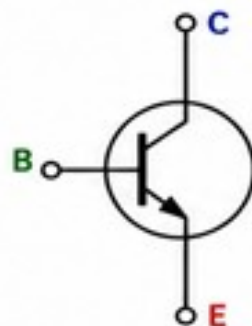
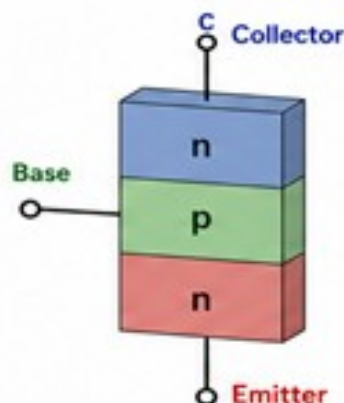
small control signal → larger controlled current

8. Transistor as a Tiny Electronic Switch



A transistor uses a very **small input** signal to control a **larger current**.
It is the basic building block of modern electronics and computers.

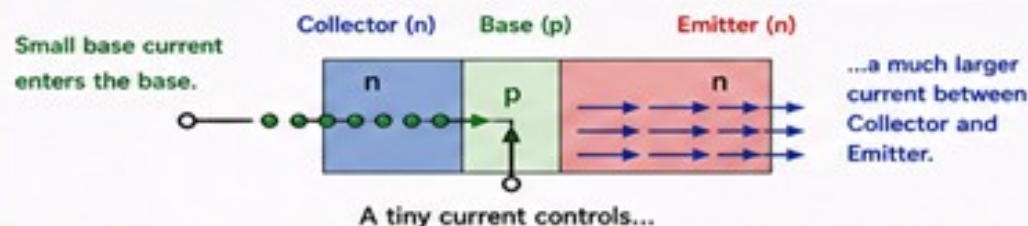
NPN Transistor – Structure and Symbol



B – Base (control)
C – Collector (output)
E – Emitter (ground)

A small current or voltage at the Base controls the current between Collector and Emitter.

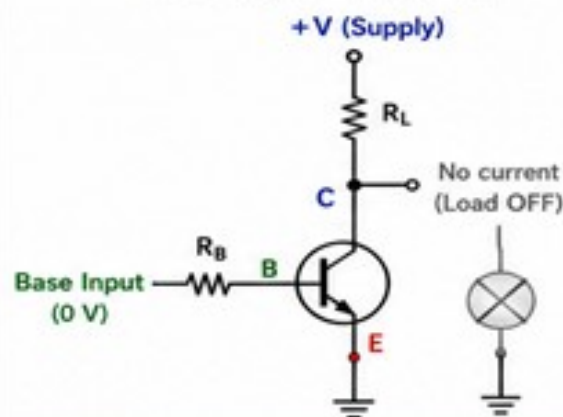
How It Works (Idea)



Operation as a Switch (NPN Transistor)

OFF (Cutoff)

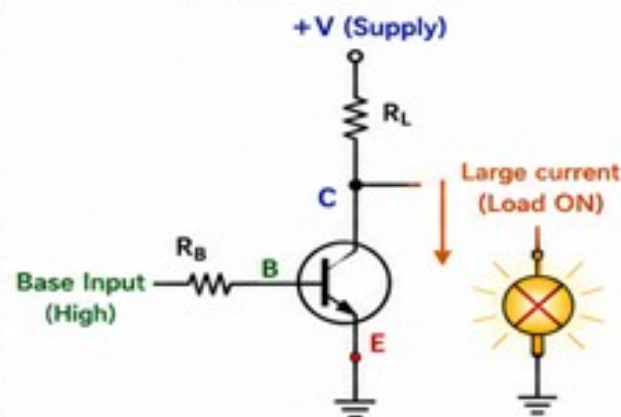
No or very small base current.



Transistor is OFF.
No current flows from Collector to Emitter.
Load is OFF.

ON (Saturation)

Sufficient base current.



Transistor is ON.
Large current flows from Collector to Emitter.
Load is ON.

Key Points

- Very **small** input controls **large** output.
- Very fast (nanoseconds).
- Very small and low power.
- No moving parts (solid-state).
- Billions are used inside modern chips.



Used In

- Computers (CPUs, memory, GPUs)
- Phones, tablets, IoT devices
- Power supplies and chargers
- Cars, appliances, almost everything



Small Input
(Base Signal)



Controls



Large Current
(Output)



Drives
Loads



Performs
Useful Work

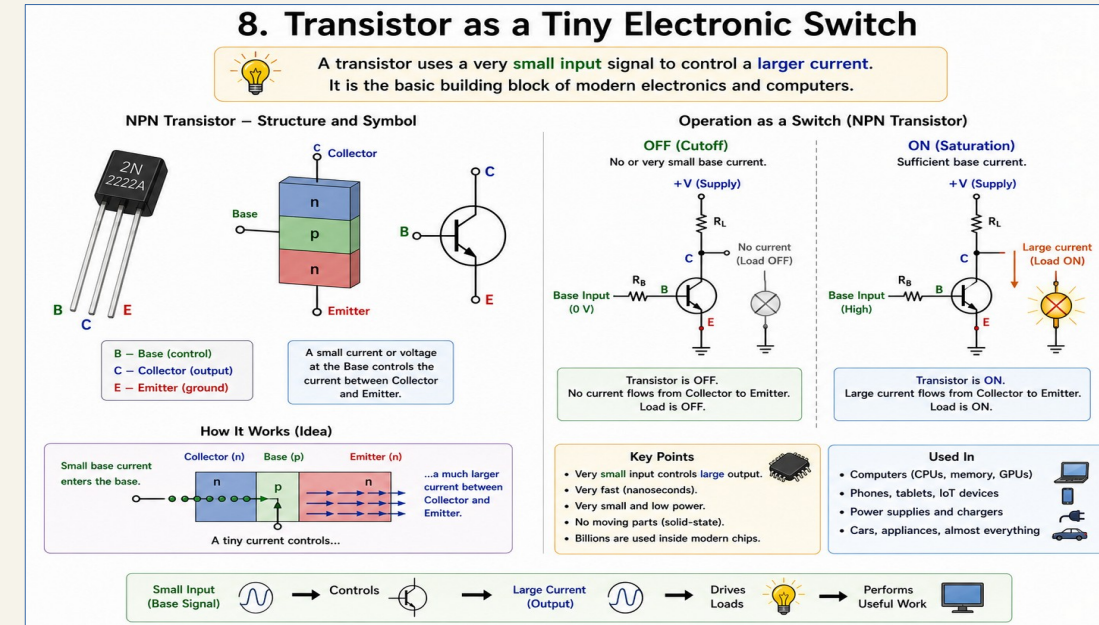


Historical note: transistor invention

A small device with huge consequences.

The first working point-contact transistor was demonstrated at Bell Labs in 1947.
The key idea was not only “smaller part.”
It was a new physical method for controlling current.
Transistors replaced many vacuum tubes because they could be smaller.

More detail: Transistors replaced many vacuum tubes because they could be smaller, cooler, and more reliable. Use this as a short history hook. The students do not need names in detail, but they should see that modern computers came from real inventions.



From transistor to integrated circuit

Many parts become one small chip.

One transistor is useful.

Many transistors can build logic gates.

Many logic gates can build registers, counters, adders, and control circuits.

The next great step was the integrated circuit: many electronic components made together on one piece of semiconductor material.

This made computers smaller and more powerful.

This is important for their future laptop lesson. A chip is not magic. It is many tiny devices built together.

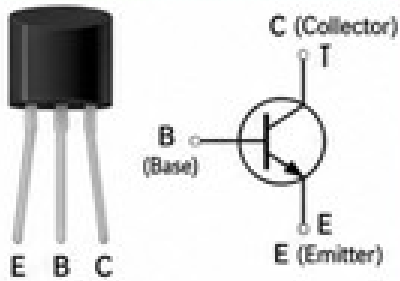
9. From Transistor to Integrated Circuit

Engineers first used single transistors to control electric signals.

Then they combined many parts to build circuits.

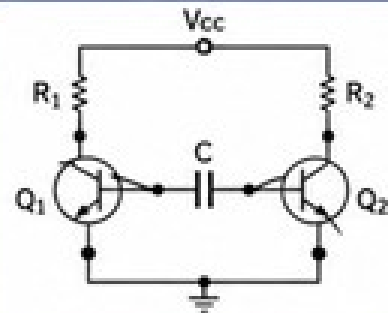
Later, they placed **thousands, millions, or even billions of components on one small semiconductor chip**. This is called an **Integrated Circuit (IC)**.

1 One Transistor



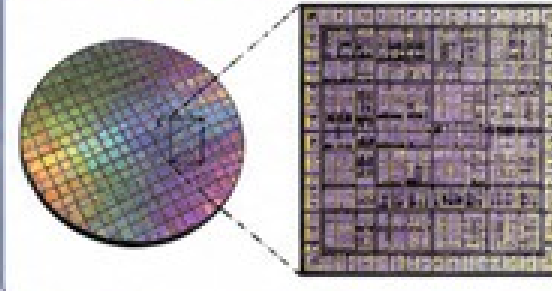
A single transistor works like a tiny switch or amplifier.

2 Small Circuit



Many parts (transistors, resistors, capacitors, etc.) are combined to do a job.

3 Many Transistors on Silicon



Thousands to billions of tiny transistors and components are built on one small silicon chip.

4 Integrated Circuit (IC)



The chip is protected in a packages with pins so it can be used in devices.

Why Integrated Circuits are Better



Smaller Size

More functions in a very small space.



Faster Signals

Short distances on the chip make signals faster.



More Reliable

Fewer wires and joints mean less chance of failure.



Less Wiring

Components are connected inside the chip.



Lower Cost per Function

Many components made together cost less.



Foundation of Modern Computers

ICs are the heart of phones, computers, cars, and more.



From one transistor to billions on a chip — this is the great invention that changed the world!

Historical note: integrated circuit

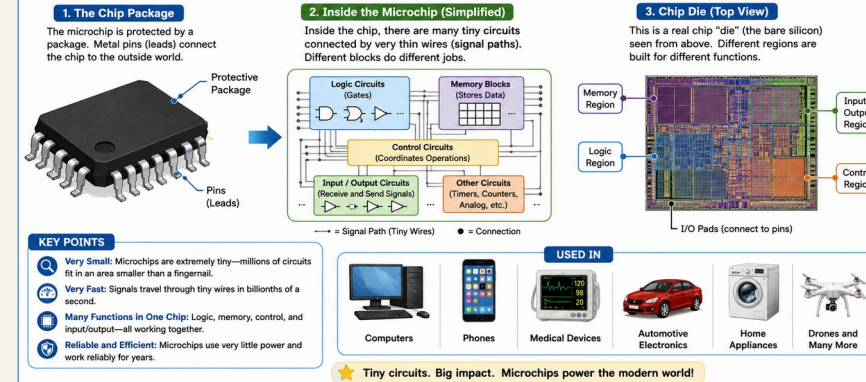
Miniaturization becomes the main road.

In the late 1950s, engineers demonstrated working integrated circuits.
The big idea:
do not wire every part separately.
Make many parts together in one small solid piece.
This reduced size, wiring problems, cost, and power use.
Modern CPUs, RAM chips, and controllers come from this road.

Computer History Museum notes Jack Kilby tested an early working integrated circuit in 1958. It is a clean historical milestone for the miniaturization story.

10. Inside a Microchip — Many Tiny Circuits

A microchip is a tiny piece of **semiconductor** material (usually **silicon**) that contains millions to billions of tiny electronic circuits. Each small circuit does a job, and together they make a powerful device.

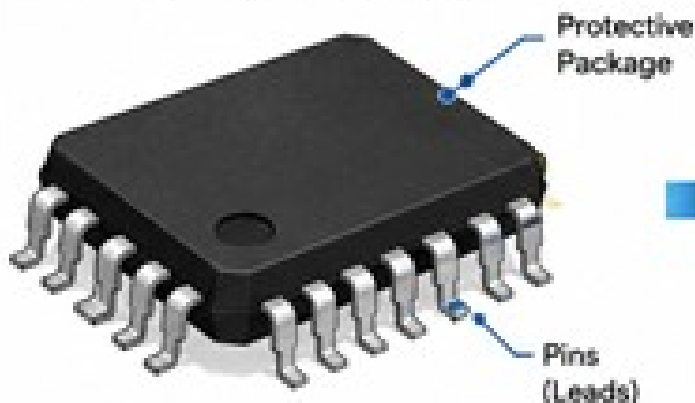


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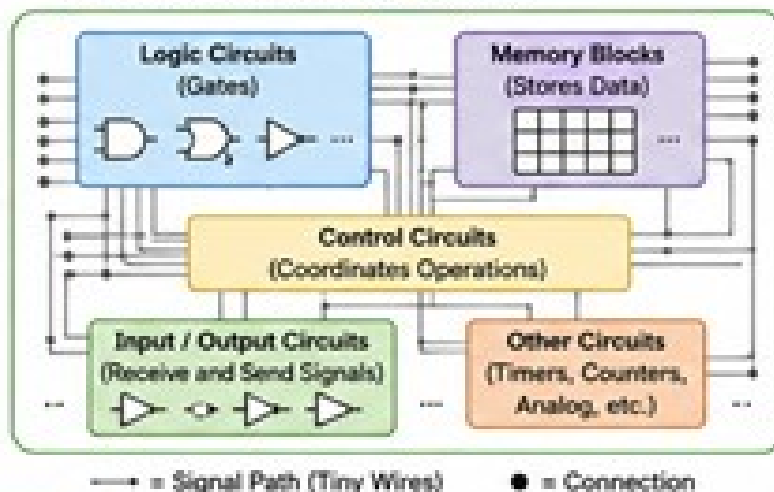
1. The Chip Package

The microchip is protected by a package. Metal pins (leads) connect the chip to the outside world.



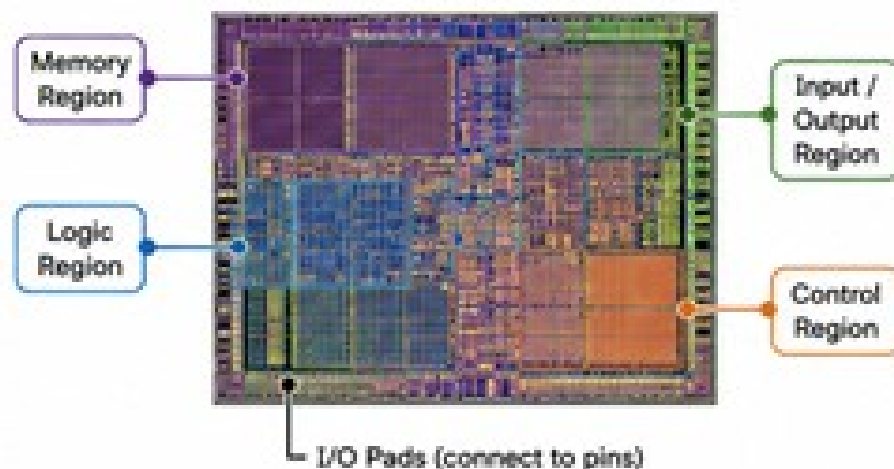
2. Inside the Microchip (Simplified)

Inside the chip, there are many tiny circuits connected by very thin wires (signal paths). Different blocks do different jobs.



3. Chip Die (Top View)

This is a real chip "die" (the bare silicon) seen from above. Different regions are built for different functions.



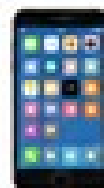
KEY POINTS

- Very Small:** Microchips are extremely tiny—millions of circuits fit in an area smaller than a fingernail.
- Very Fast:** Signals travel through tiny wires in billionths of a second.
- Many Functions in One Chip:** Logic, memory, control, and input/output—all working together.
- Reliable and Efficient:** Microchips use very little power and work reliably for years.

USED IN



Computers



Phones



Medical Devices



Automotive Electronics



Home Appliances



Drones and Many More

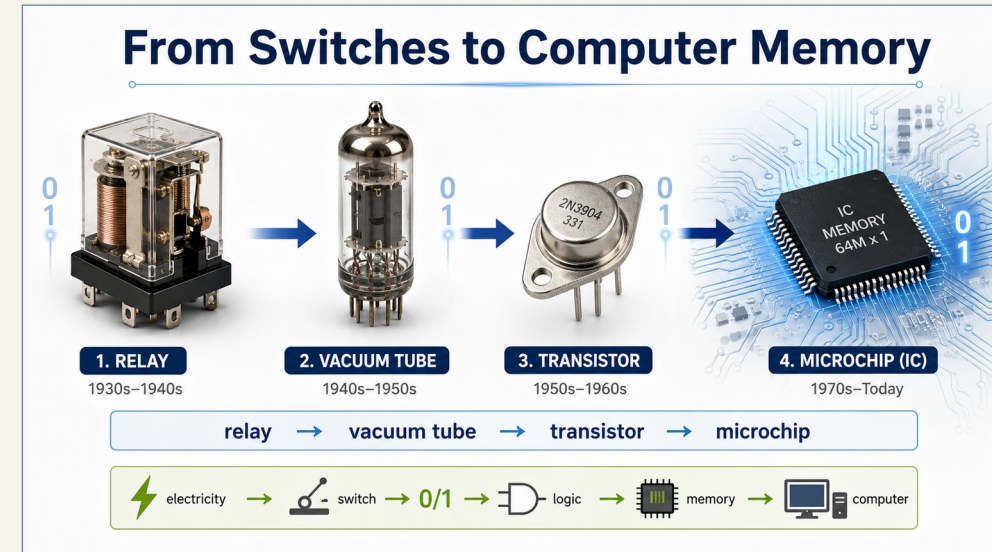


Tiny circuits. Big impact. Microchips power the modern world!

Main history chain

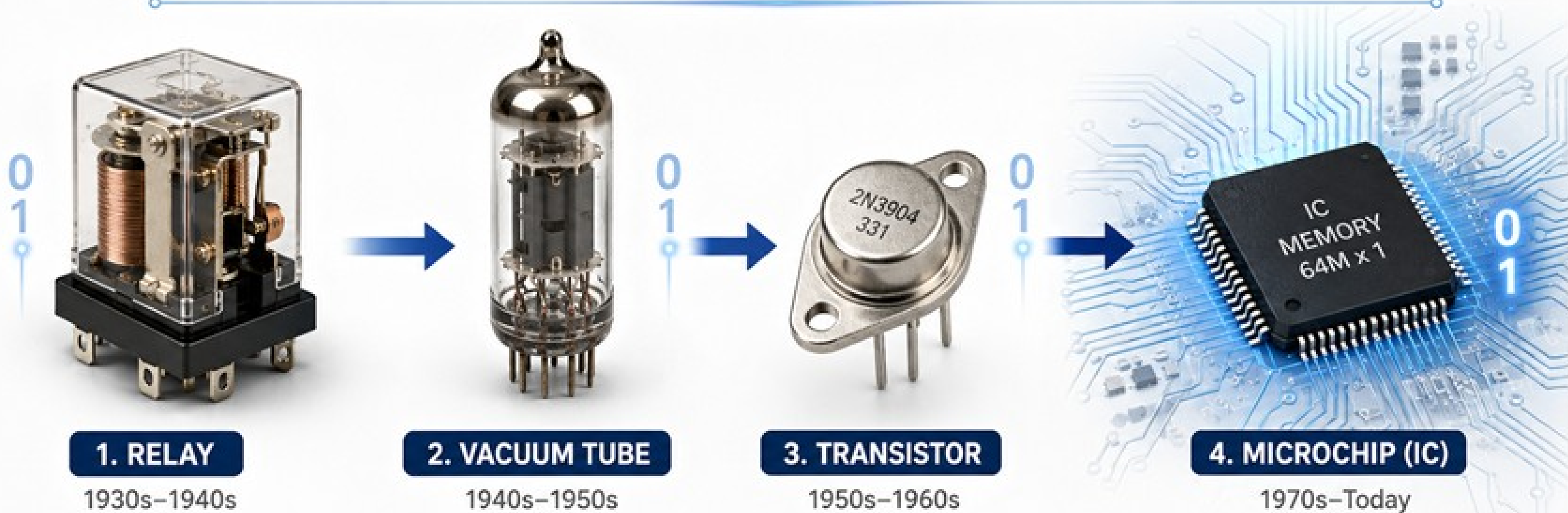
Same idea, better physical implementation.

Mechanical switch: human or mechanical motion controls a circuit.
Relay: electromagnet moves a contact.
Vacuum tube: electric field controls electrons in vacuum.
Transistor: semiconductor controls current.
Integrated circuit: many tiny controlled switches in one chip.
The idea stays similar:
control current to represent and process information.

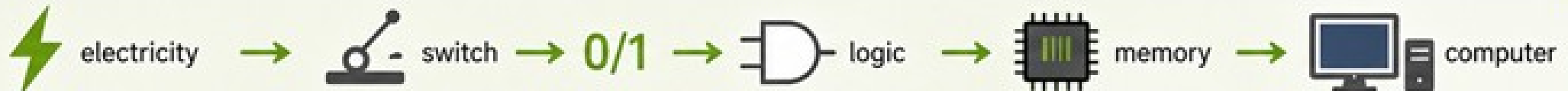


This is the central summary of the lecture. It is exactly the mix of physics and history: every generation uses a different physical mechanism for the same computing need.

From Switches to Computer Memory



relay → vacuum tube → transistor → microchip



Part 5 — Calculation is not enough

A computer also needs memory.

Logic can calculate.
But a useful
computer must also
remember.
It must remember:
numbers for
calculation
instructions of a
program
letters and symbols
images and sound

35. How Data is Stored in Computers

All data in a computer is stored as 0 and 1.

Letter A	0100	0001
Number 5	0000	0101
Image (pixel)	 0	 1

Text, numbers, pictures, sound, video –

Memory as physical state

Remembering means keeping a state for later.

A computer bit must be stored physically.
Different technologies can store a bit in different ways:
charge or no charge
magnetic direction
trapped electrons
flip-flop circuit state
Today we focus on one important idea:
charged capacitor = 1
not charged capacitor = 0

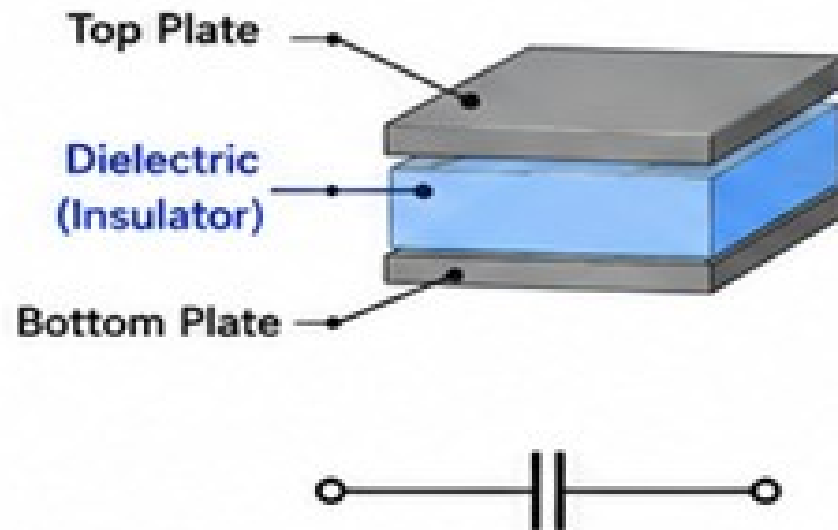
More detail: capacitor stores charge.

This prevents the oversimplification that all memory is only a capacitor. DRAM uses capacitors, but SSD stores data differently. We will compare them later.

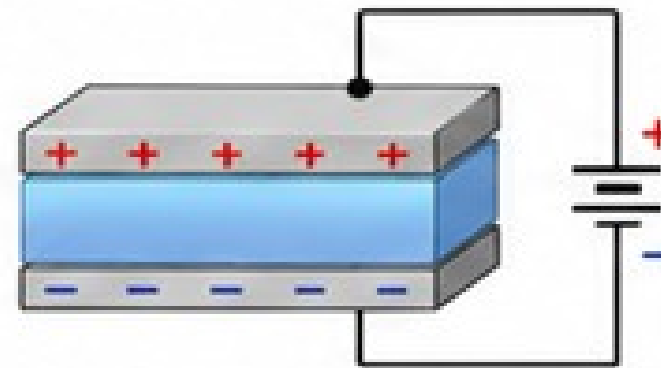
13. Capacitor Stores Charge: Charged = 1, Empty = 0

A capacitor can store electrical charge. We can use it to represent **1 (charged)** or **0 (empty)**. This is the basic idea behind memory.

How a Capacitor Works



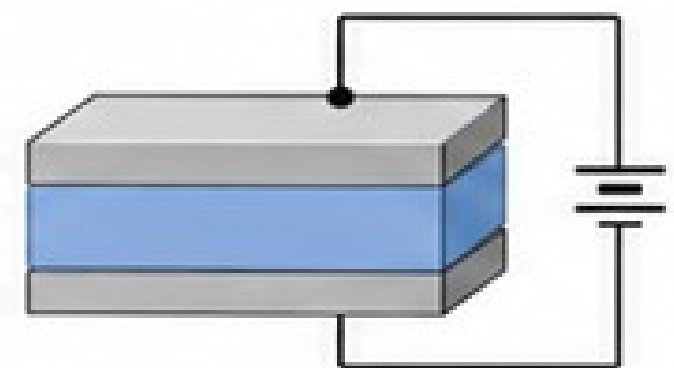
Charged = 1



Charge is stored.
We read this as 1.

Voltage across plates = **High**

Empty = 0



No charge stored.
We read this as 0.

Voltage across plates = **Low**



Capacitors can store charge very quickly and are very small.
Billions of capacitors are used in modern memory chips.

Capacitor: the needed physics

A small device that stores electric charge.

A capacitor usually has two conductive plates separated by an insulator.

When voltage is applied, charge can build up on the plates.

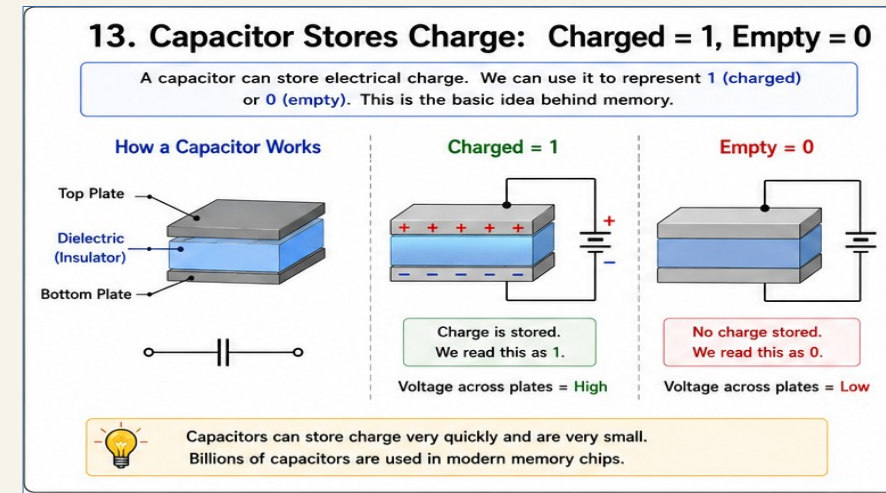
The capacitor stores electrical energy in an electric field.

Simple memory idea:

charged capacitor = 1

not charged capacitor = 0

This is where capacity/ capacitance belongs. It is not random physics anymore. It solves the memory problem.



Capacitance formula

Optional but useful physics.

Capacitance tells how much charge a capacitor can store for a given voltage.

$$C = Q / V$$

C = capacitance

Q = stored charge

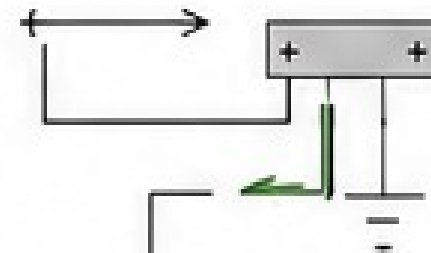
V = voltage

Bigger capacitance means more charge can be stored at the same voltage.

For this lecture, the concept is more important than calculation.

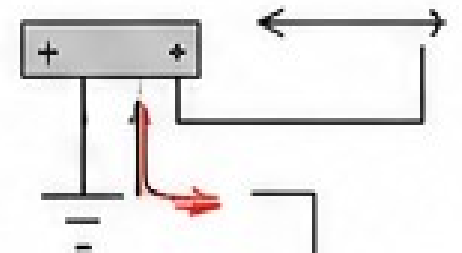
38. Charge and Discharge in a Capacitor

Charging (store 1)



Current flows in
Charge builds up
on the capacitor

Discharging (becomes 0)



Charge leaves
the capacitor

Keep the formula, but do not make the lecture about calculations. Students should remember: capacitor stores charge; capacitance is storage ability.

Capacitor is not a battery

Small storage, different purpose.

A capacitor can store charge quickly.

It can also release charge quickly.

But it usually stores much less energy than a battery.

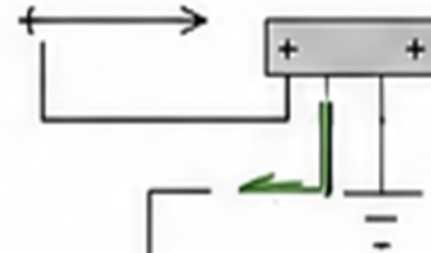
Battery: chemical energy, longer storage, power source.

Capacitor: electrical charge storage, fast charging and discharging.

In computers, capacitors help with memory, filtering, timing, and

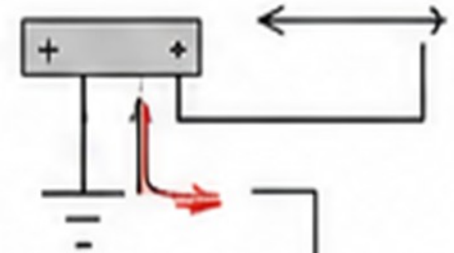
38. Charge and Discharge in a Capacitor

Charging (store 1)



Current flows in
Charge builds up
on the capacitor

Discharging (becomes 0)



Charge leaves
the capacitor

This prevents a common misunderstanding. Capacitor is storage, but not the same kind of storage as a phone battery.

DRAM: memory from tiny capacitors

One bit can be stored as charge.

DRAM means Dynamic Random Access Memory.

Modern DRAM memory cells are based on a tiny capacitor and a transistor.

The capacitor stores charge.

The transistor helps read or write the bit.

charged = 1

not charged = 0

This is a direct link between physics and computer memory.

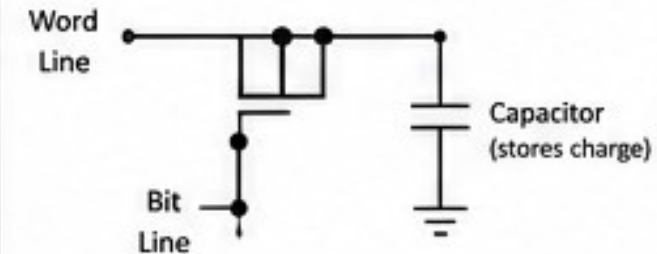
37. RAM Cell (Simplified)

A DRAM cell stores 1 bit.

It uses one transistor and one capacitor.

Capacitor stores charge.

Transistor controls access.



Charged = 1
Not charged = 0

Charge leaks away.
So DRAM must refresh.

IBM credits Robert Dennard with the DRAM invention, patented in 1968. The important teaching point is not the patent. It is the physical idea: charge becomes a stored bit.

Why DRAM is “dynamic”

Charge slowly leaks away.

A capacitor does not hold charge forever.
Charge slowly leaks away.
If the computer does nothing,
the stored 1 can become
unclear or lost.
So DRAM must refresh
memory again and again.
Refresh means:
read the stored state and
write it again before it
disappears.

This is a very important idea. Memory is not magic. It is maintenance of physical states. If power is gone, RAM loses data.

39. DRAM Refresh Cycle

Charge leaks away with time.
So the memory controller refreshes it.



Refresh happens thousands of times per second.
If refresh stops – data is lost.

RAM needs power

Working memory disappears when power is off.

RAM is fast working memory. The CPU uses RAM while the computer is working. Open programs use RAM. Temporary data uses RAM. But normal RAM needs power. When power is off, the data in RAM is lost. This is why unsaved work can disappear after shutdown or power failure.

Connect this to real student experience: if a computer crashes and the file was not saved, RAM did not keep it.

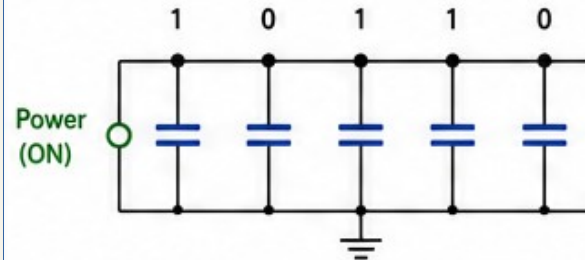
14. RAM Loses Data When Power Is Off

RAM (Random Access Memory) uses capacitors to store data.

When power is removed, the charge leaks away and the data is lost.

Power ON

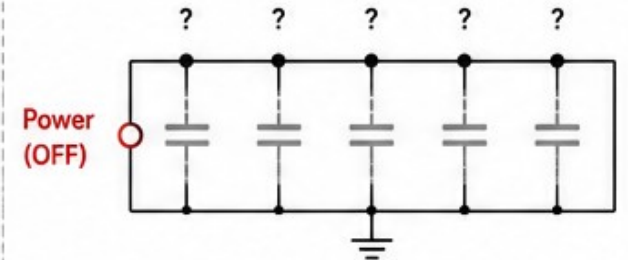
Capacitors are charged.
Data is stored and can be read.



Data example: 1 0 1 1 0

Power OFF

Capacitors discharge (leak).
Data is lost.



Data is lost: ? ? ? ? ?



RAM is fast and easy to change, but it needs power all the time to keep data.

SSD is different

Long-term storage keeps data without power.

SSD means Solid State Drive.
SSD stores files and programs
for later.

It keeps data when the power is
off.

SSD is not the same as RAM.

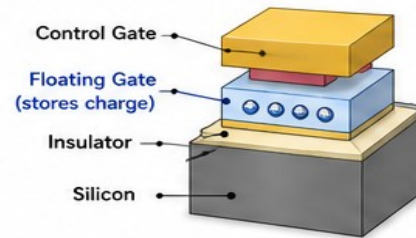
RAM: fast working area,
temporary, needs power.

SSD: long-term storage, keeps
files, slower than RAM but
persistent.

15. SSD Keeps Data Without Power

SSDs (Solid State Drives) use flash memory.
Flash memory stores charge in a special way that does not leak quickly.
So data stays even when power is off.

Inside Flash Memory Cell



Charged = 1

Charge is trapped on
floating gate.



Data is kept
(until we erase it).

Empty = 0

No charge on
floating gate.



Data is kept
(until we program it).



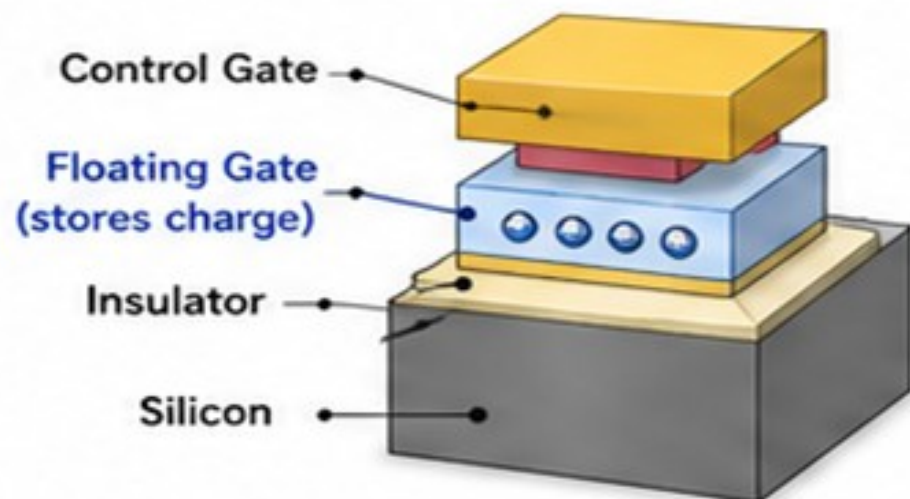
SSDs keep data even without power, so they are perfect for long-term storage
(like files, photos, documents, operating system, etc.).

This prepares the practical laptop lesson. Many students confuse memory and storage. Teach this early.

15. SSD Keeps Data Without Power

SSDs (Solid State Drives) use flash memory.
Flash memory stores charge in a special way that does not leak quickly.
So data stays even when power is off.

Inside Flash Memory Cell



Charged = 1

Charge is trapped on floating gate.



Data is kept
(until we erase it).

Empty = 0

No charge on floating gate.



Data is kept
(until we program it).



SSDs keep data even without power, so they are perfect for long-term storage (like files, photos, documents, operating system, etc.).

CPU, RAM, SSD: three roles

Do not mix them.

CPU calculates and controls.
RAM holds data while the computer is working.
SSD keeps files and programs when the computer is off.
Example:
Open a hospital information system.
Program files come from SSD.
Working patient data is held in RAM.

More detail: CPU follows instructions and processes data. All three are needed.
Use a hospital information system example. This makes the hardware relevant to medical IT support.

17. CPU / RAM / SSD — Roles in a Hospital Information System

In a hospital information system, all parts work together to store, process, and provide patient data quickly and safely.

CPU (Processor)



- The brain of the system.
- Runs programs.
- Processes data and makes decisions.

Example:

Doctor opens a patient record.
CPU processes the request.

RAM (Memory)



- Working memory.
- Holds data and programs that are being used right now.
- Very fast, but temporary.

Example:

Patient data is loaded into RAM
while the doctor is viewing it.

SSD (Storage)



- Long-term storage.
- Saves all data permanently even when power is off.
- Slower than RAM, but large.

Example:

Patient records, images, and
reports are saved on the SSD.



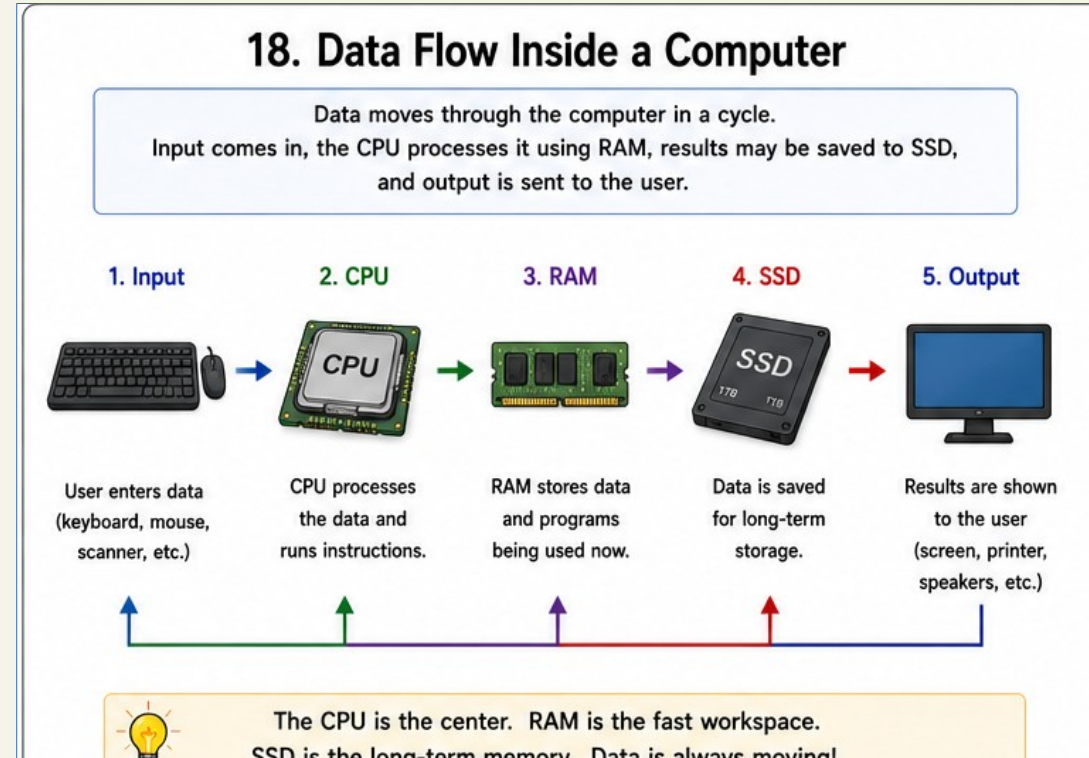
CPU thinks and decides. RAM works quickly. SSD remembers forever.
Together they make the hospital system fast, reliable, and safe.

Part 6 — From hardware to software

Why computer logic matters for software users.

Software looks like buttons, menus, forms, and screens.
But under the screen there are instructions.
Instructions are processed by logic circuits.
Logic circuits are built from physical switches.
So when we learn computer logic, we learn how software can follow exact rules.
This helps us understand errors, settings, and troubleshooting.

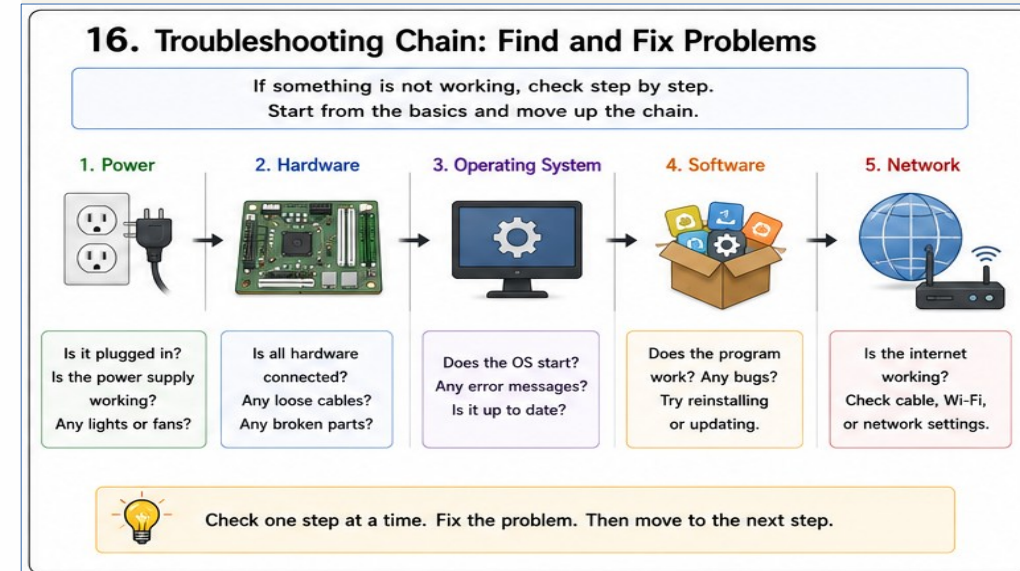
This slide directly explains why a future software user should understand computer logic. It is not only for engineers.



Thinking like a computer

Exact rules, exact states, exact order.

A computer does not guess like a human.
It follows exact rules.
Important computer thinking:
state: 0 or 1, ON or OFF,
connected or not connected
condition: IF this, THEN that
sequence: step 1, step 2, step 3
memory: keep data or lose data



More detail: error: find where the chain breaks This is useful for software, networks, and hospital systems. This connects to previous algorithm/flowchart lessons. Computer logic is not only electronics. It is a style of thinking useful for support work.

16. Troubleshooting Chain: Find and Fix Problems

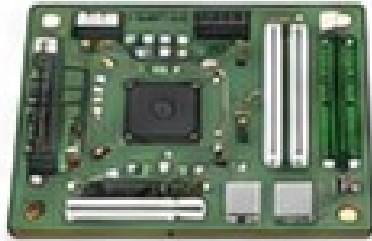
If something is not working, check step by step.
Start from the basics and move up the chain.

1. Power



Is it plugged in?
Is the power supply working?
Any lights or fans?

2. Hardware



Is all hardware connected?
Any loose cables?
Any broken parts?

3. Operating System



Does the OS start?
Any error messages?
Is it up to date?

4. Software



Does the program work? Any bugs?
Try reinstalling or updating.

5. Network



Is the internet working?
Check cable, Wi-Fi, or network settings.



Check one step at a time. Fix the problem. Then move to the next step.

Troubleshooting view

Every computer problem is a broken chain somewhere.

If a computer does not work, do not guess first.

Find the chain:

power → motherboard →
CPU/RAM/SSD → operating system
→ software → network → user action

A broken cable is physical.

Bad RAM is memory hardware.

A wrong setting is software logic.

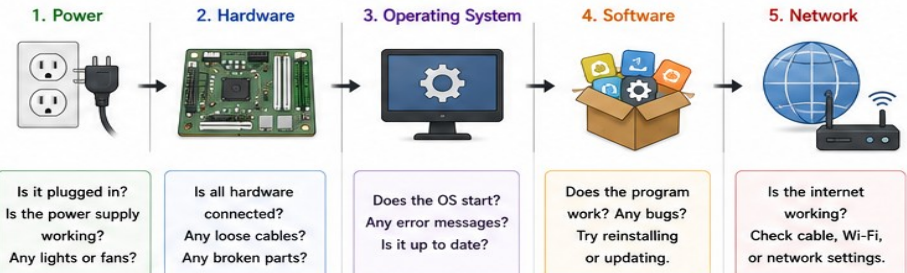
No network is a communication problem.

More detail: Good IT support finds the weak link step by step.

This gives practical value for medical IT. They are not going to design CPUs, but they must report and troubleshoot safely.

16. Troubleshooting Chain: Find and Fix Problems

If something is not working, check step by step.
Start from the basics and move up the chain.



Check one step at a time. Fix the problem. Then move to the next step.

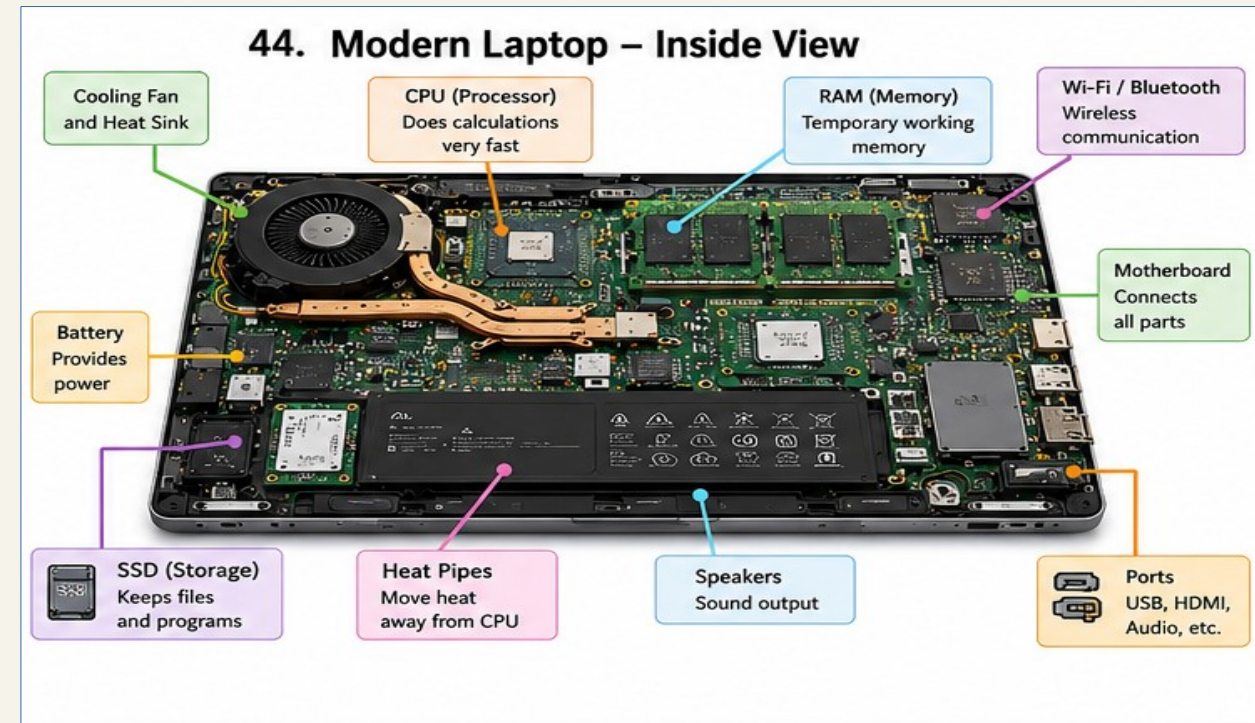
Practical lesson preview

Next time: open a real laptop.

Next lesson we can look inside a real laptop or PC. We will identify:

- battery or power input
- motherboard
- CPU and cooling system
- RAM
- SSD

More detail: ports and cables fan and airflow Then we connect each part to today's logic. This is the reward. After history and physics, students get a real machine. The lecture should make laptop parts meaningful, not random.



Class / online activity 1

Switch states and truth tables.

Task A: Write the state.

Switch open = ?

Switch closed = ?

Task B: Complete AND.

0 AND 0 = ?

0 AND 1 = ?

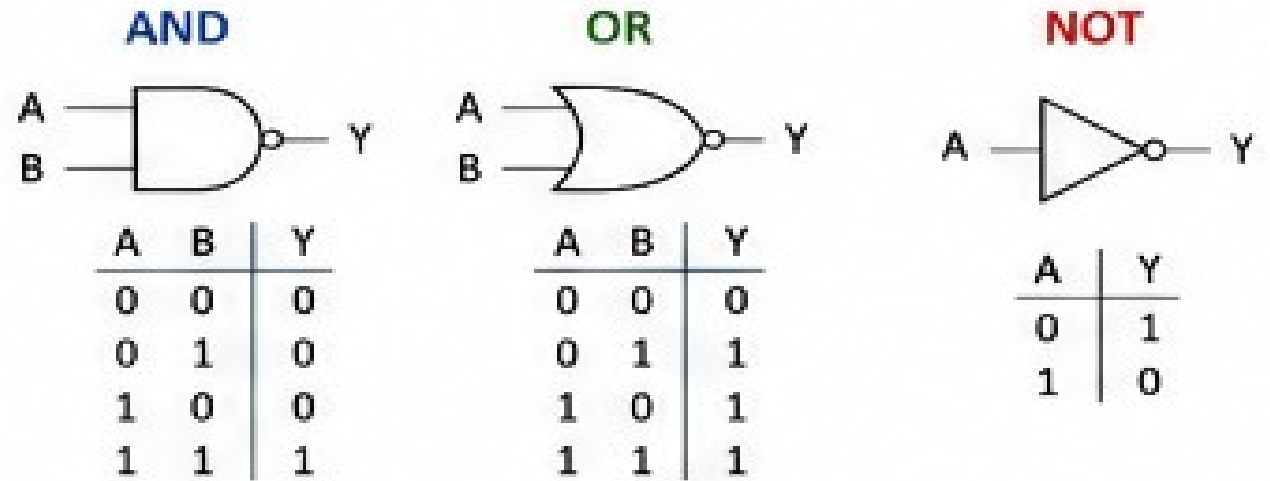
1 AND 0 = ?

33. Logic Gates – The Building Blocks

Computers use simple logic gates.

These gates work with 0 and 1.

They are made from transistors.



More detail: 1 AND 1 = ? Task C: Complete OR and NOT.

This can be used online. Students answer in chat, notebook, or worksheet. Keep it short and repetitive.

Class / online activity 2

Find the technology chain.

Choose one device:
relay, vacuum tube, transistor,
capacitor, RAM, SSD.

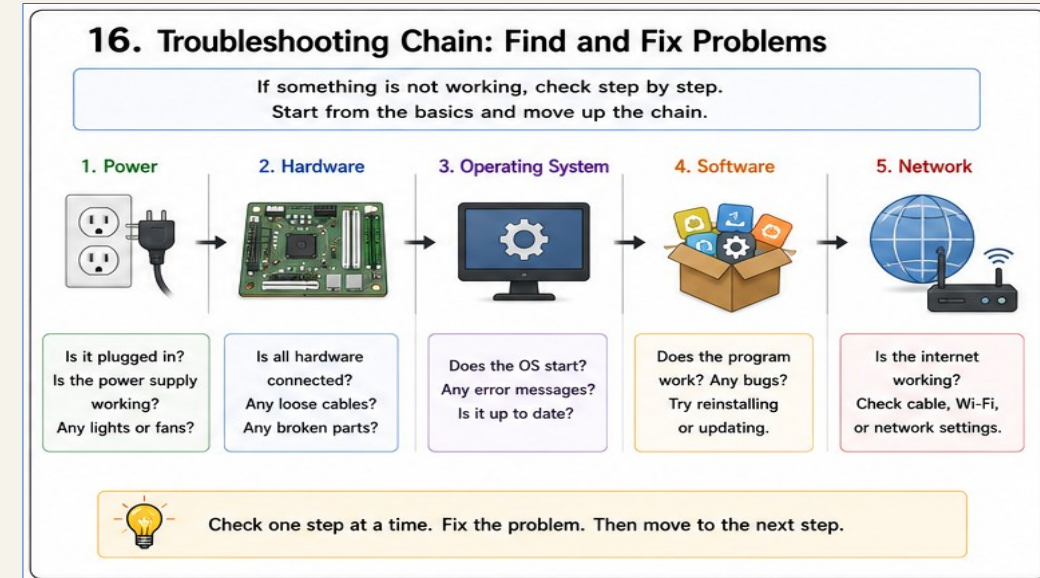
Fill this table:

Input:

Physical process:

Output:

Why useful for computers?

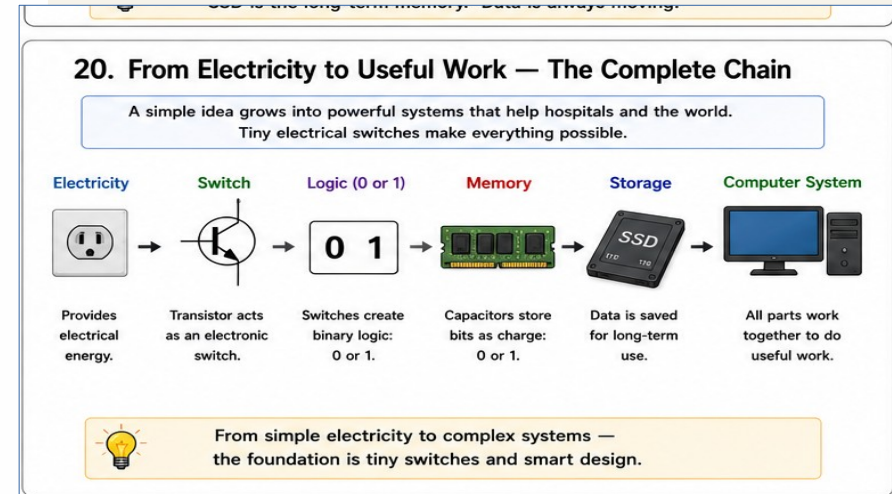


More detail: Example: Relay Input: small current Process: electromagnet moves contact Output: another circuit turns ON
This activity directly uses the course method from Lecture 8: input → process → output. It also prepares them for equipment troubleshooting.

Mini quiz

Short review questions.

1. What are the two states of a switch?
2. What number can mean OFF?
3. What number can mean ON?
4. What device uses electromagnetism to move a contact?
5. Why were relays too slow for later computers?
6. What did vacuum tubes control?
7. What replaced many vacuum tubes?



More detail: 8. What stores charge? 9. Why does DRAM refresh? 10. What is the difference between RAM and SSD? Use as oral review. For weak students, allow one-word answers. The goal is recall of the chain, not perfect English.

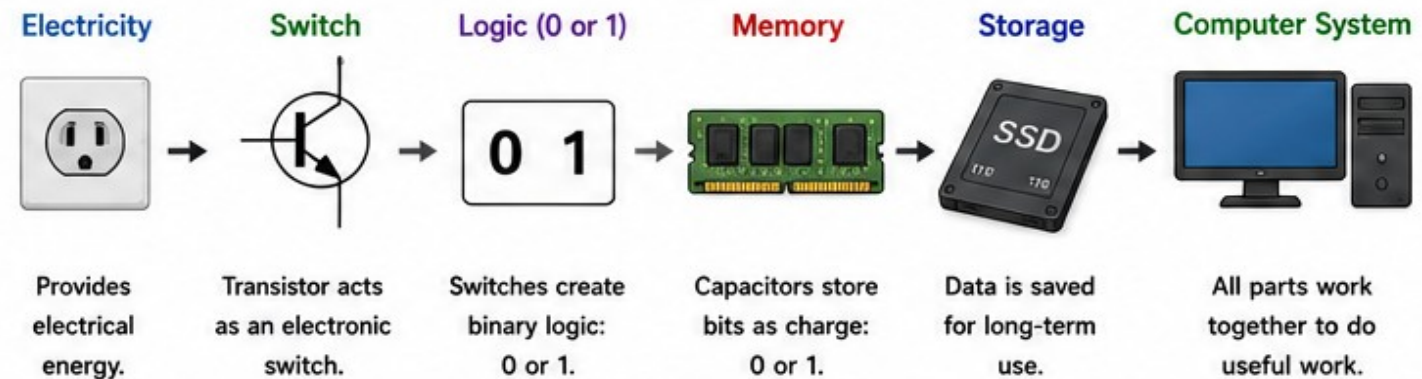
Mini quiz answers

Open after trying.

1. OFF and ON.
2. 0.
3. 1.
4. Relay.
5. Relays move metal parts; movement is slow and parts wear out.
6. Electron flow in a tube.
7. Transistors.

20. From Electricity to Useful Work — The Complete Chain

A simple idea grows into powerful systems that help hospitals and the world.
Tiny electrical switches make everything possible.



From simple electricity to complex systems —
the foundation is tiny switches and smart design.

More detail: 8. Capacitor. 9. Because charge leaks away. 10. RAM is fast working memory and needs power. SSD keeps files without power.
This is for online self-study. Hidden answer can be used later when the slides become a web page.

Final summary

From electricity to computer.

Main chain:

electricity → switch → 0/1 →
logic → memory → computer

Technology chain:

relay → vacuum tube →
transistor → integrated circuit

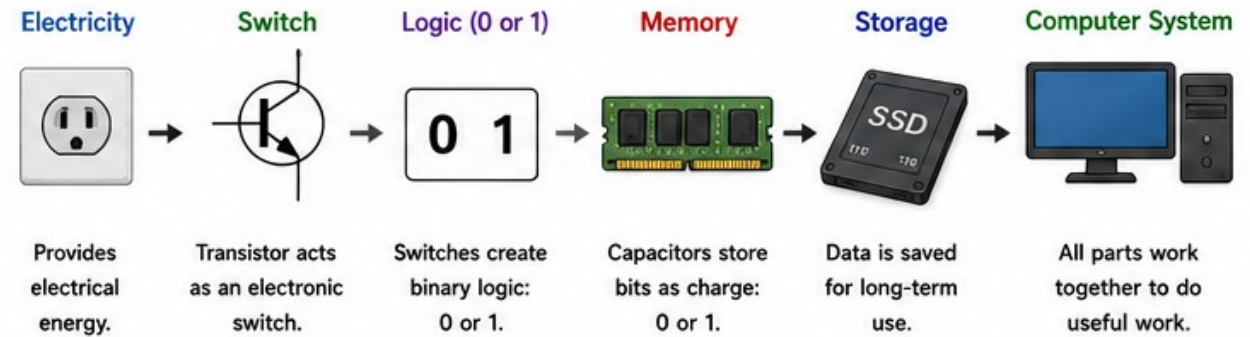
Memory chain:

charge → capacitor → bit →
RAM

Big idea:

20. From Electricity to Useful Work — The Complete Chain

A simple idea grows into powerful systems that help hospitals and the world.
Tiny electrical switches make everything possible.



From simple electricity to complex systems —
the foundation is tiny switches and smart design.

More detail: Computers are built from physical devices that control electrical states.
This slide should be strong and simple. It is the message of the whole lecture.

Source notes

Historical references used in the lecture.

Computer History Museum: Zuse Z3 used relays and binary arithmetic; ENIAC used almost 18,000 tubes.
Nobel Prize / Bell Labs history: Bardeen, Brattain and Shockley demonstrated the transistor in 1947 and received the 1956 Nobel Prize in Physics.
Computer History Museum / Smithsonian: early integrated circuits were demonstrated in 1958.
IBM history: Robert Dennard and IBM were issued the DRAM patent in 1968.

More detail: Computer History Museum: Zuse Z3 used relays and binary arithmetic; ENIAC used almost 18,000 tubes and had tube maintenance problems. Nobel Prize / Bell Labs history: Bardeen, Brattain and Shockley demonstrated the transistor in 1947 and received the 1956 Nobel Prize in Physics.

This slide is for teacher/source transparency. It can be hidden in classroom presentation if needed, but useful for web/source control.